EclipseSource

What every developer should know about Eclipse 4

Jonas Helming
EclipseSource

Preparation

- If you did not receive a mail before, please get one of the USB Sticks and follow the instructions.pdf in the e4 tutorial folder!
- If you followed the instructions in the mail, you are ready to go…

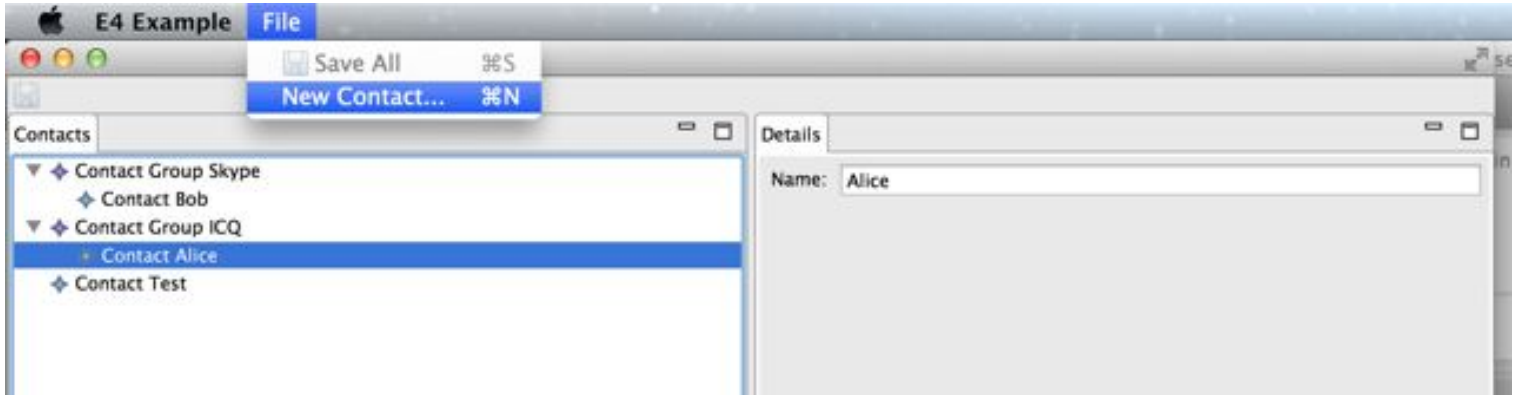
Outline

- The Application Model
- Implementing Views
- Selection Service

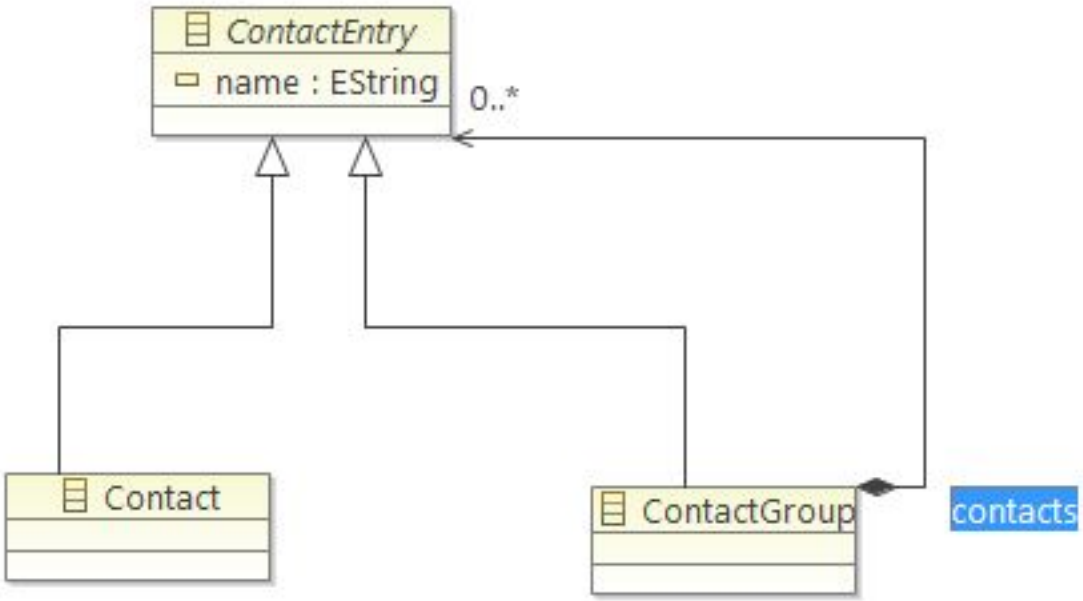
- Handlers, Commands and Items
- Editors

- Dependency Injection Details
- Modularity
- Migration

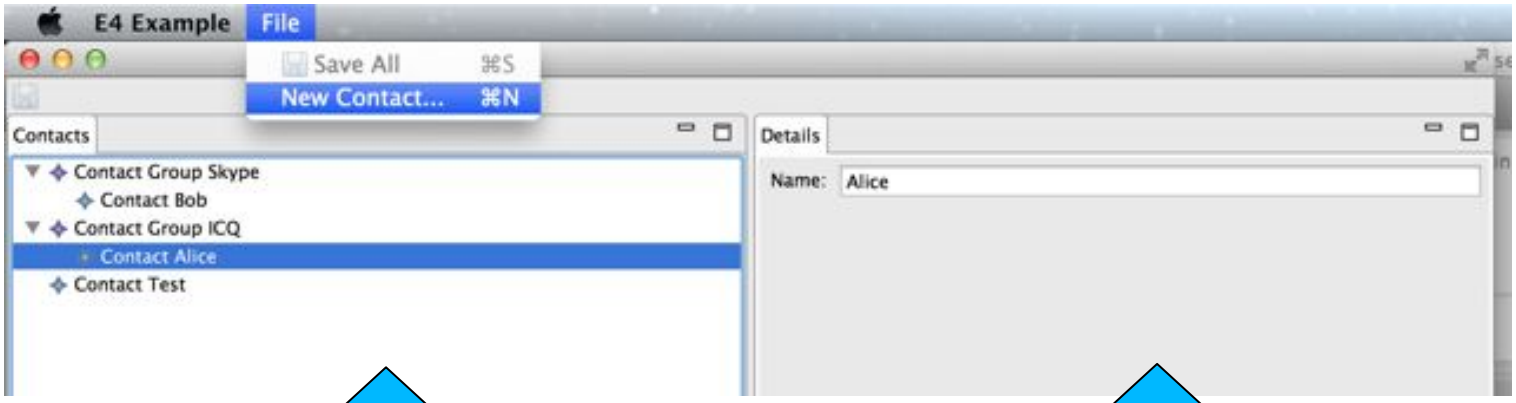
Example Application



The Application entity model



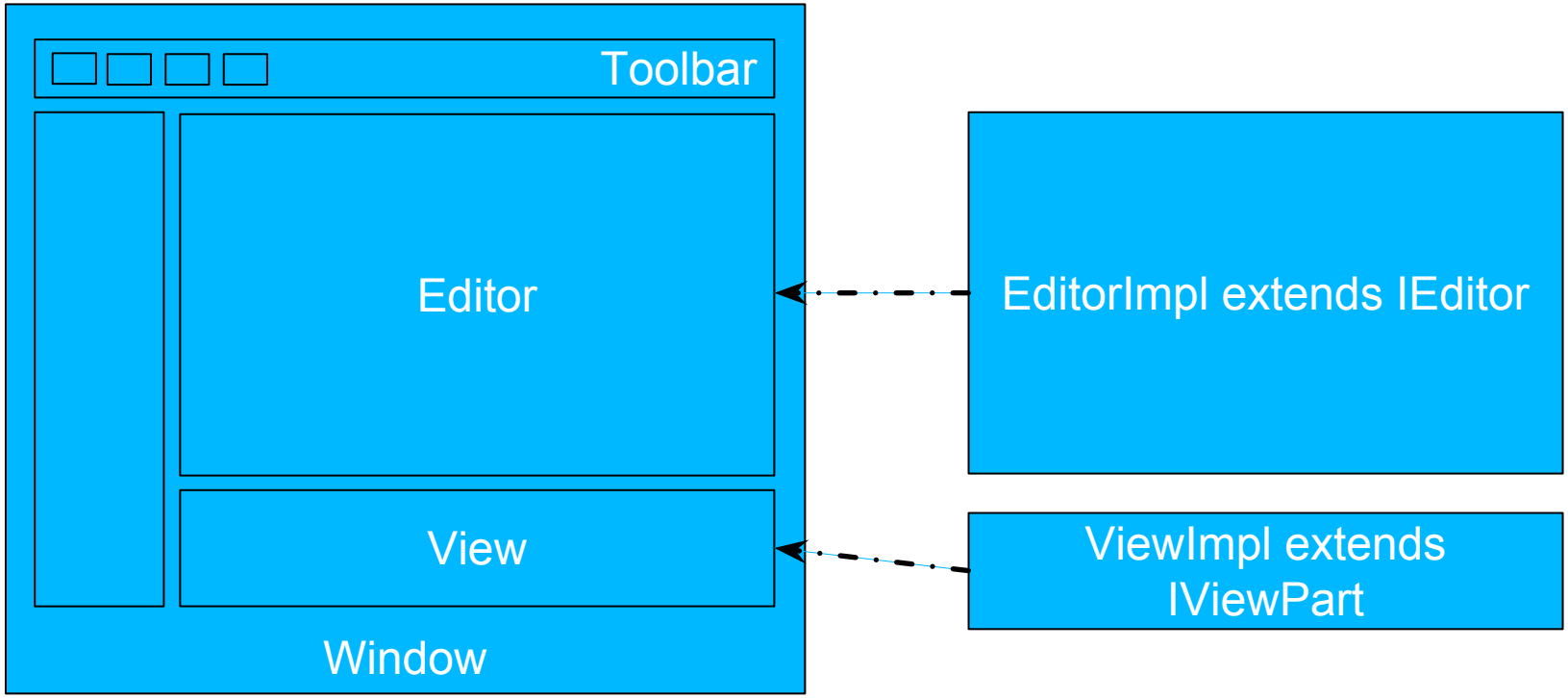
Goal 1: Two Views



ContactsView

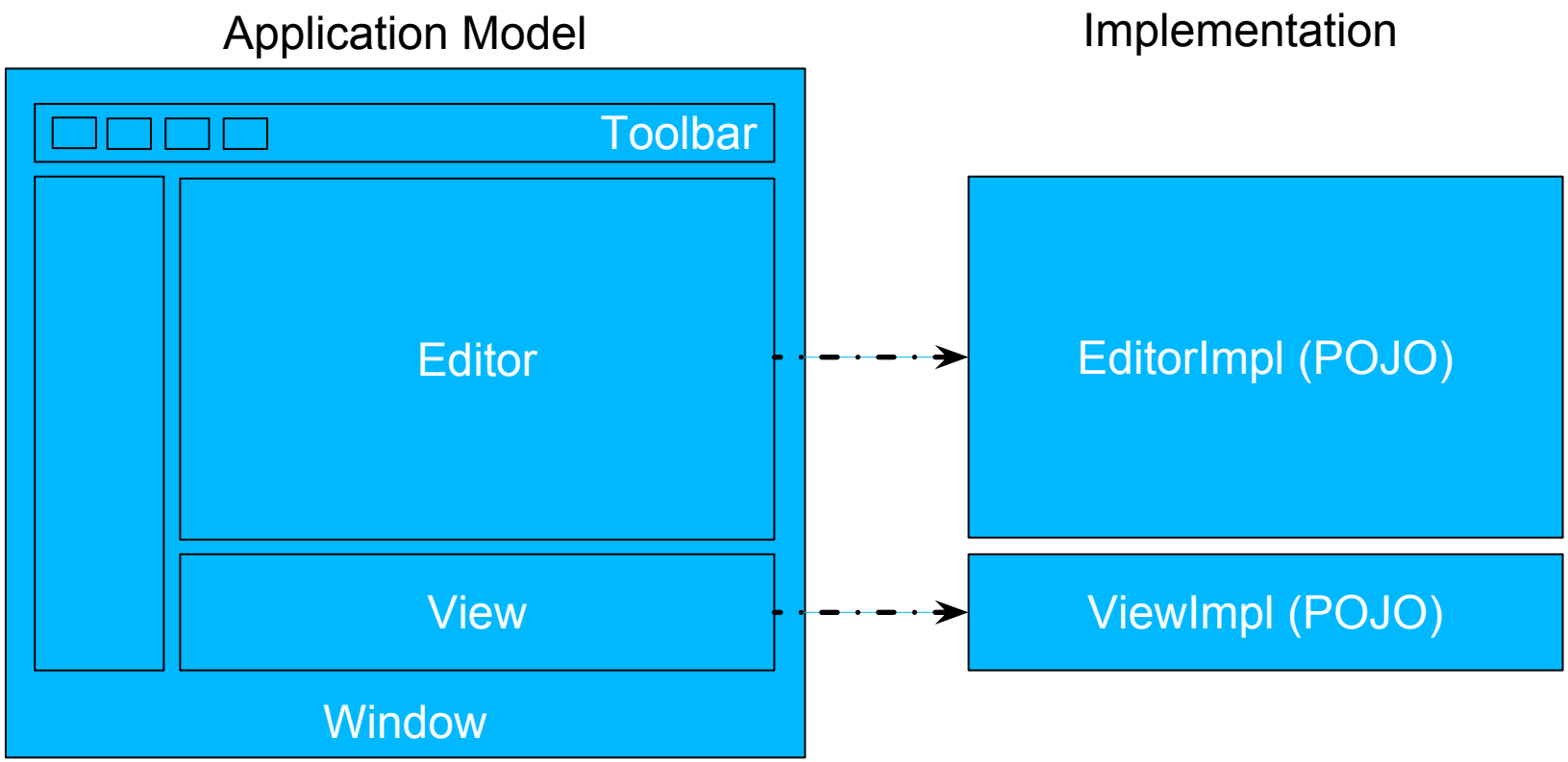
DetailsView

The 3.x Workbench



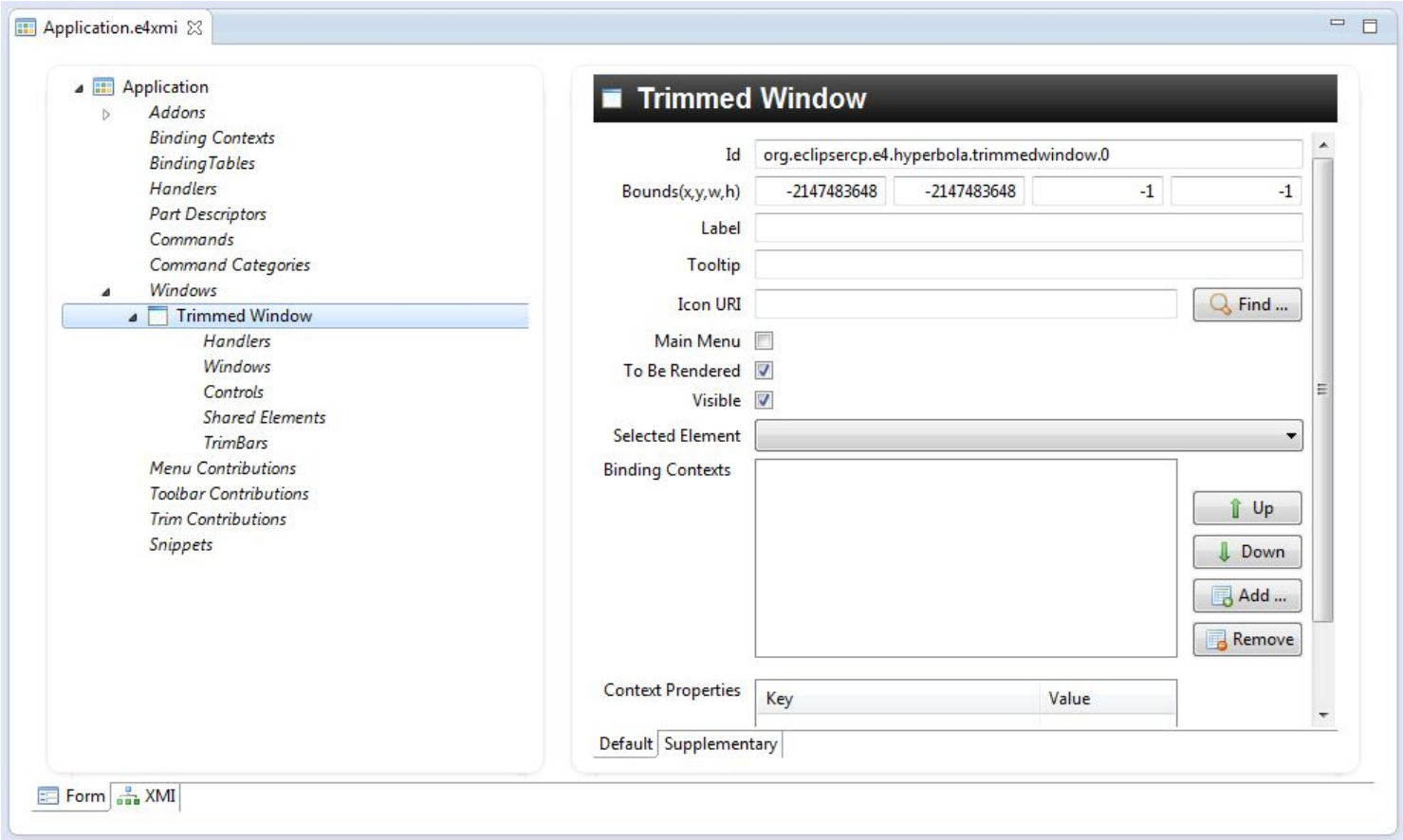
=> Strong coupling between Workbench and Implementation

The Eclipse 4 Workbench



=> Loose coupling between Workbench and Implementation

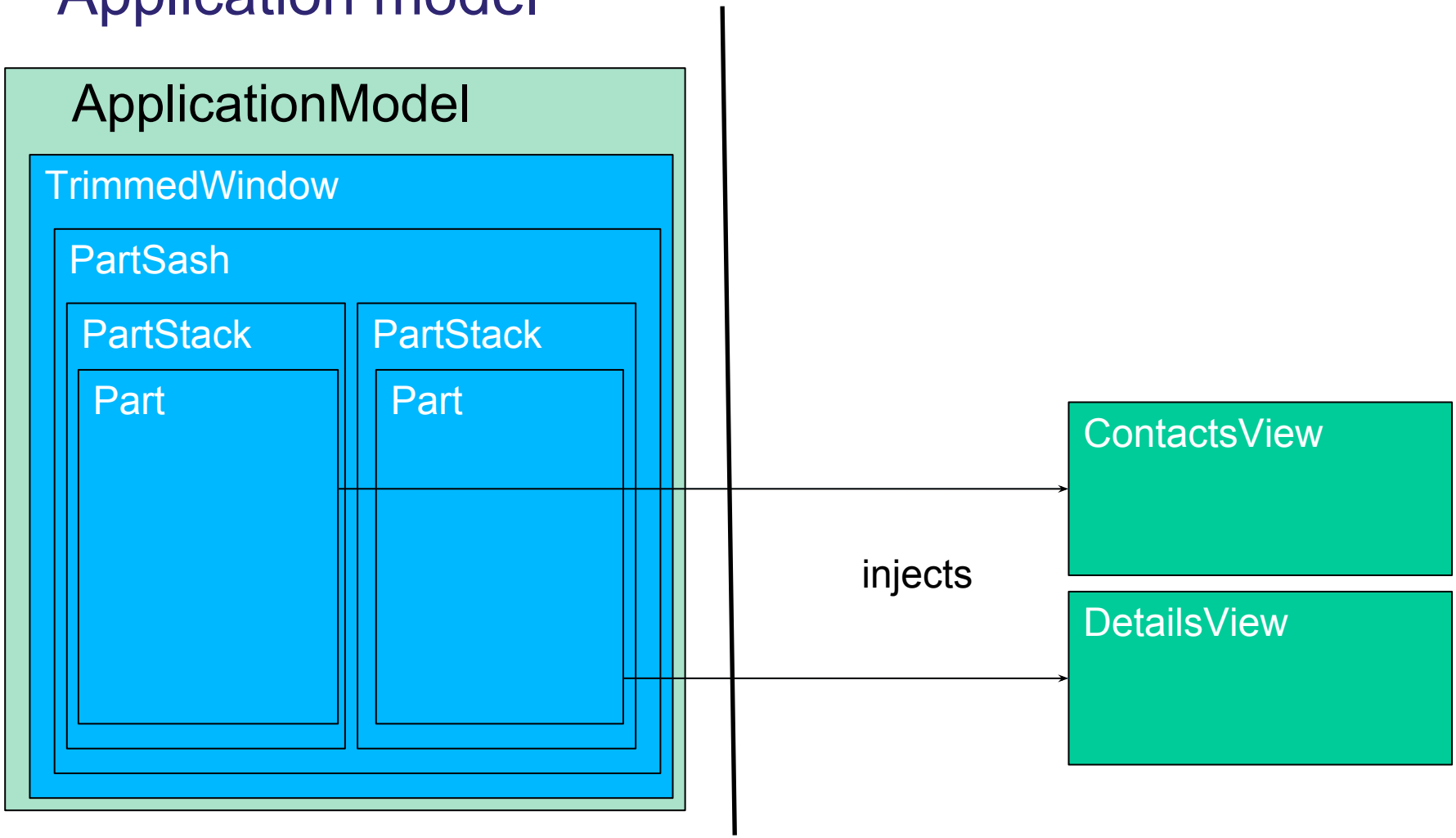
The e4 tools editor



Elements of the Application Model

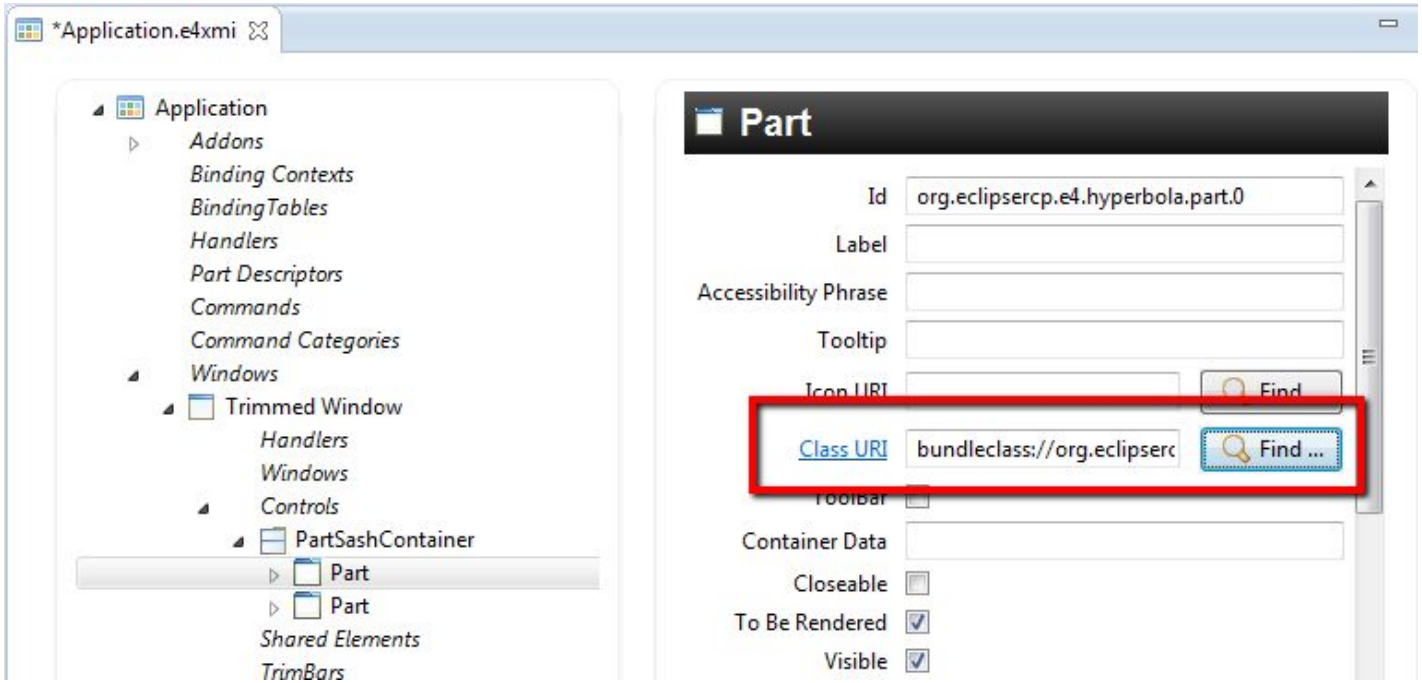
- Parts
- Windows, PartSash, PartStacks
- Handler, Commands, MenuItems, ToolBarItems
- KeyBindings
-
- Perspectives
- Placeholder
- PartDescriptors
- (…)

Application model

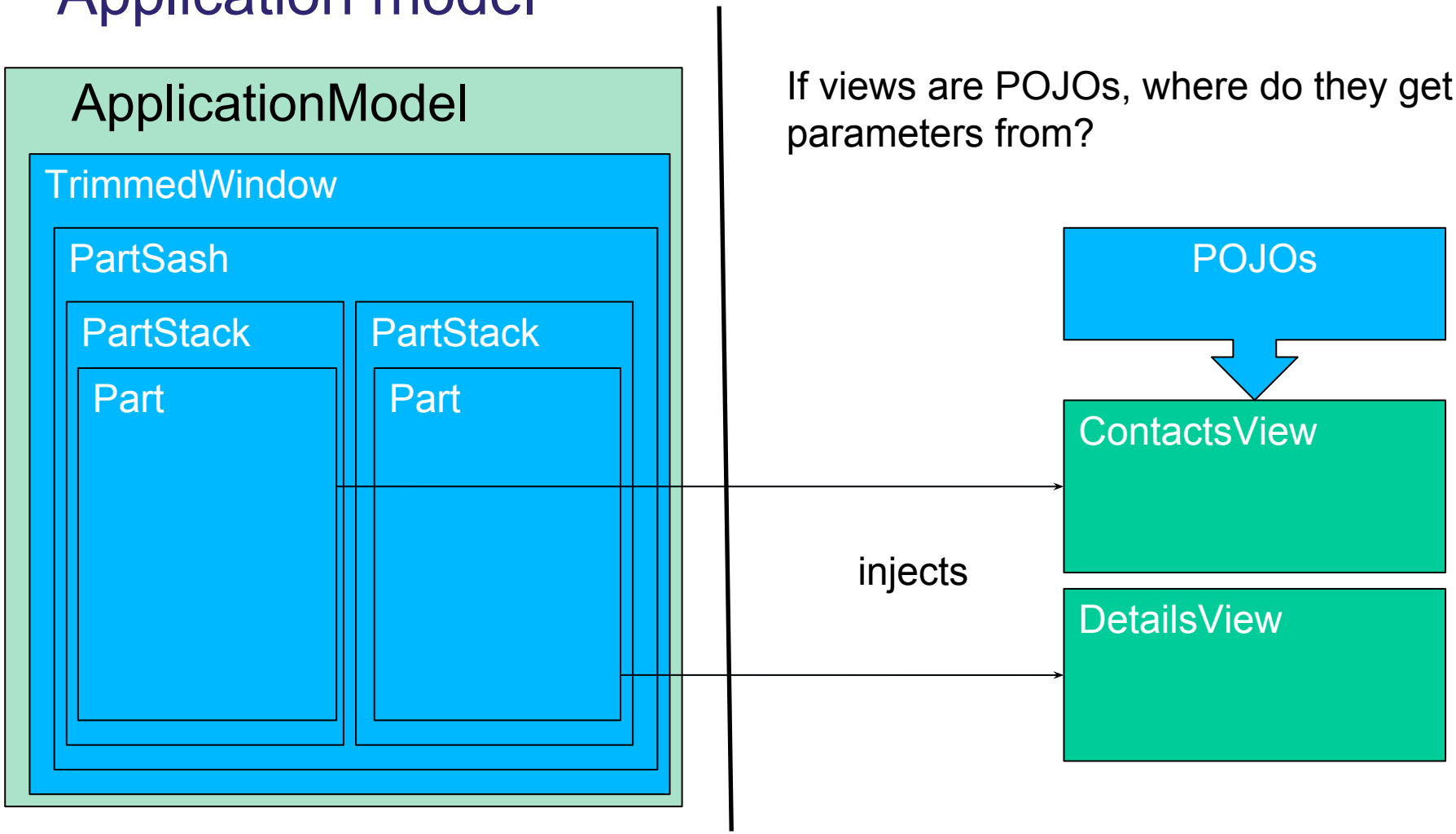


Connect views to the model

The class URI attribute binds a Part to its implementation

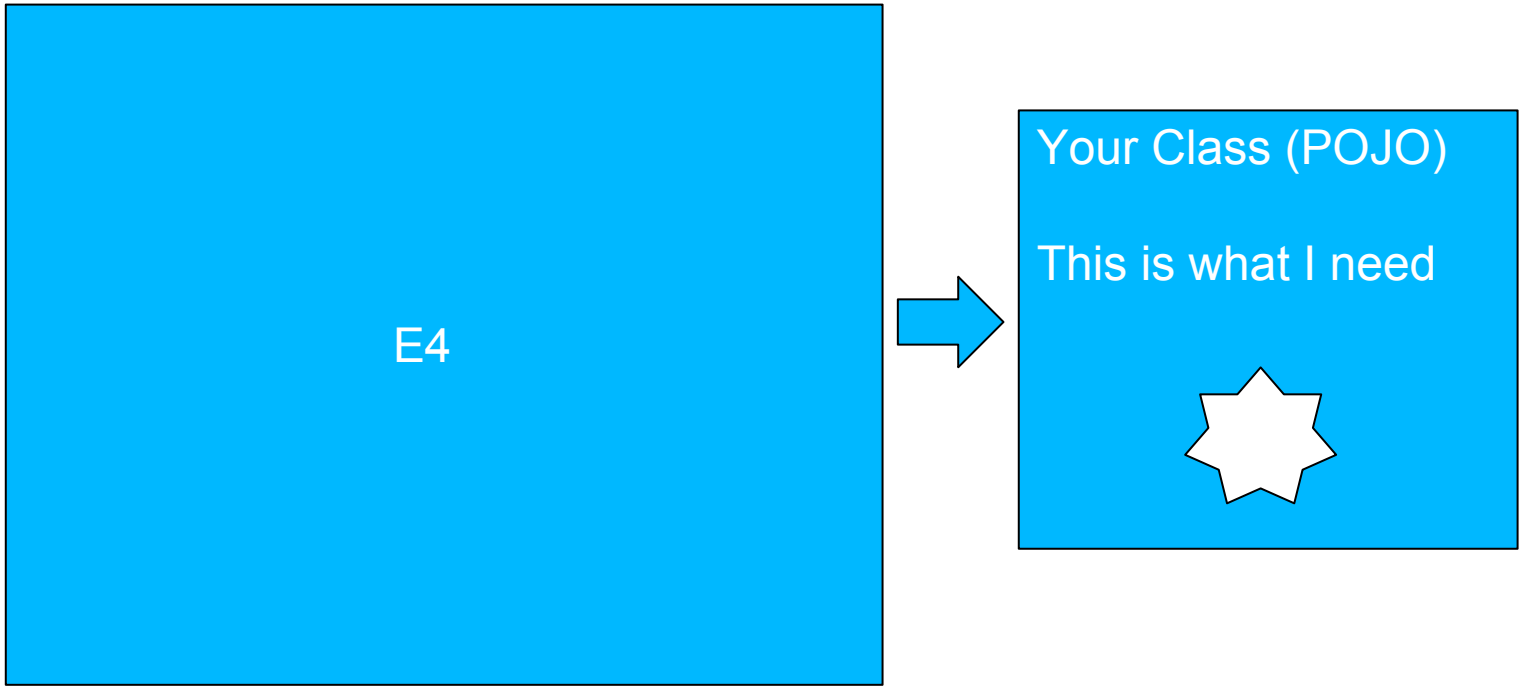


Application model



Dependency Injection

Hollywood principle: „Don‘t call us, we call you!“



What can be injected

- Elements related to the current application model element, e.g. the parent composite
- Application model elements, e.g. the main window
- Services
- Products of services, e.g. the active selection

- Own objects

- and many more...

How can Objects be injected?

1. Constructor:

```
@Inject
public void MyClass(Composite parent)
```

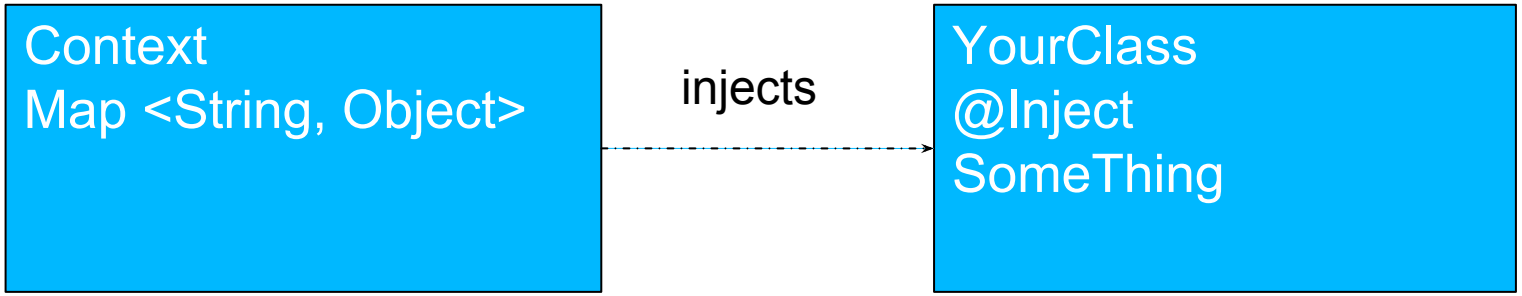
2. Fields:

```
@Inject
Service service
```

3. Methods:

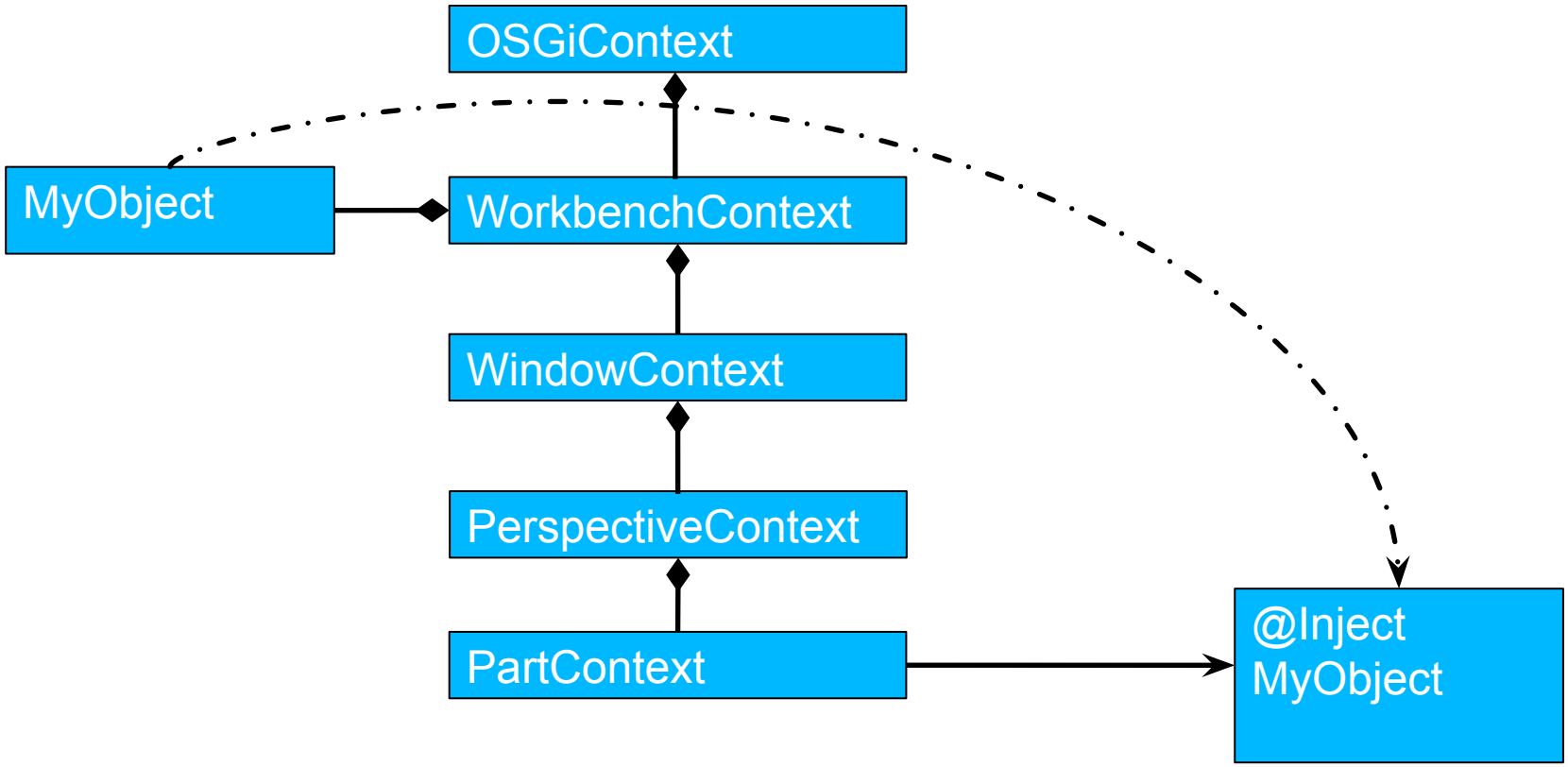
```
@Inject
public void myMethod(SomeClass class)
```

Who injects objects?



Injections are updated if the value changes
Contexts are hierachical
@Inject: Tries to satisfy the requested type
@Named: Define a key, which is used for look-up
@Optional: Injects null, if not available

Hierachy of contexts



Behaviour Annotations

- @Inject, @Named and @Optional specify, what is injected, not when

- There is often a need for a „init“ and „dispose“ method
- Custom components, such as views need to be notified, e.g. if they receive the focus

Available Behaviour Annotations

@PostConstruct: After an object is created
@PreDestroy: Before an object is destroyed
@Focus: When a UI element is focused, Views must forward the focus to a SWT widget

@Execute: Called to execute a handler
@CanExecute: Checks if a handler can be executed

All annotations include @Inject, therefore, parameters get injected

Views in Eclipse 4

- No need to implement any interface
- Start to design the view as you would do it without knowing about Eclipse
- Needed parameters are injected later on, e.g. the parent composite:
- @Inject
  MyView(Composite parent){
  //Implement View
- Services are injected, too, e.g.:
  @Inject
  ContactService contactService

Annotations in Views

```
public class ContactsView {

    @Inject
    ContactService contactService;

    @Inject
    ESelectionService selectionService;


    @Inject
    public ContactsView(Composite parent) {

    }

    @PostConstruct
    public void init() {


    }

    @Focus
    public void setFocus(){

    }

    @PreDestroy
    public void dispose(){

    }

}
```

Services

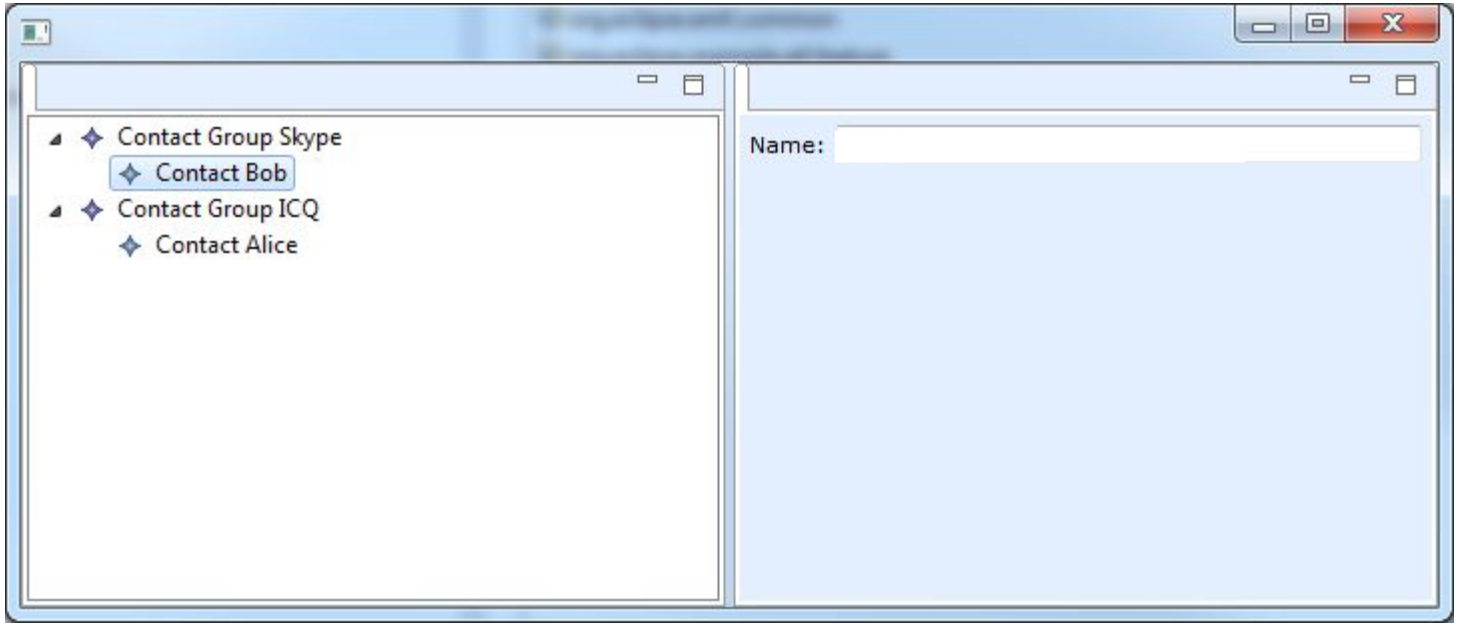
Create UI

Init UI

Set Focus

Dispose UI

Result task 1



SWT Helper in model.edit

SWTExampleHelper.**createTreeViewer**(Composite): creates a TreeViewer with provider

SWTExampleHelper.**createTextWithLabel**(Composite); to create a Label showing „Name:“ and a SWT Text

SWTExampleHelper.**dispose**(TreeViewer)

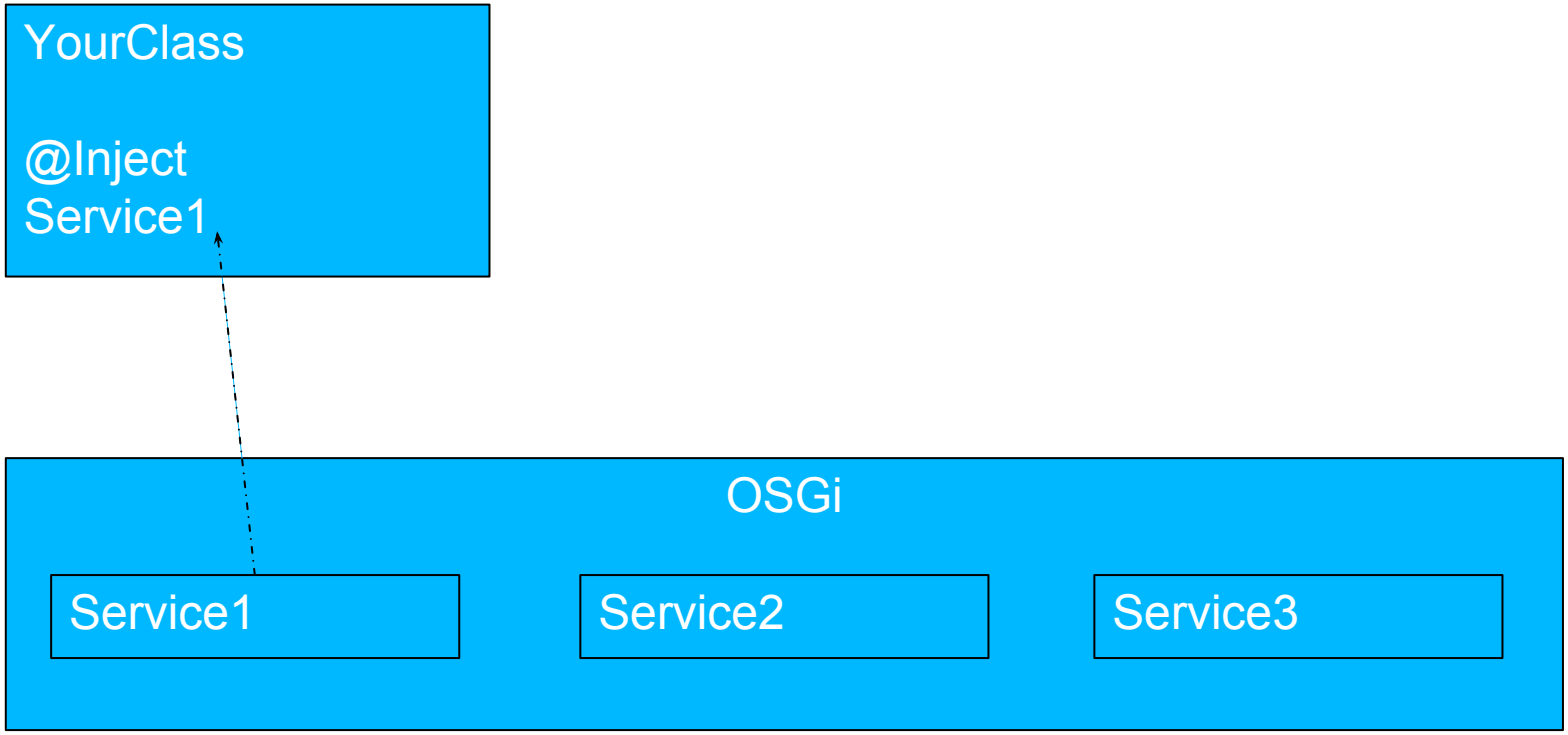
Task: Application Model

- Open the Application model (Application.e4xmi)
- Add a PartSashContainer to the window
- Add two PartStacks and Parts within them
- Implement two POJOs for the parts (use SWTExampleHelper)
  - Left: ContactsView, Right: DetailsView
  - Use the ContactService to retrieve an input for the TreeViewer (in a @PostConstruct method)
  - treeViewer.setInput(contactService.getInput());
- Connect the parts to the POJOs (class URI)
- *Optional:*
  - *Create different layouts*
  - *Change labels of parts and the window*

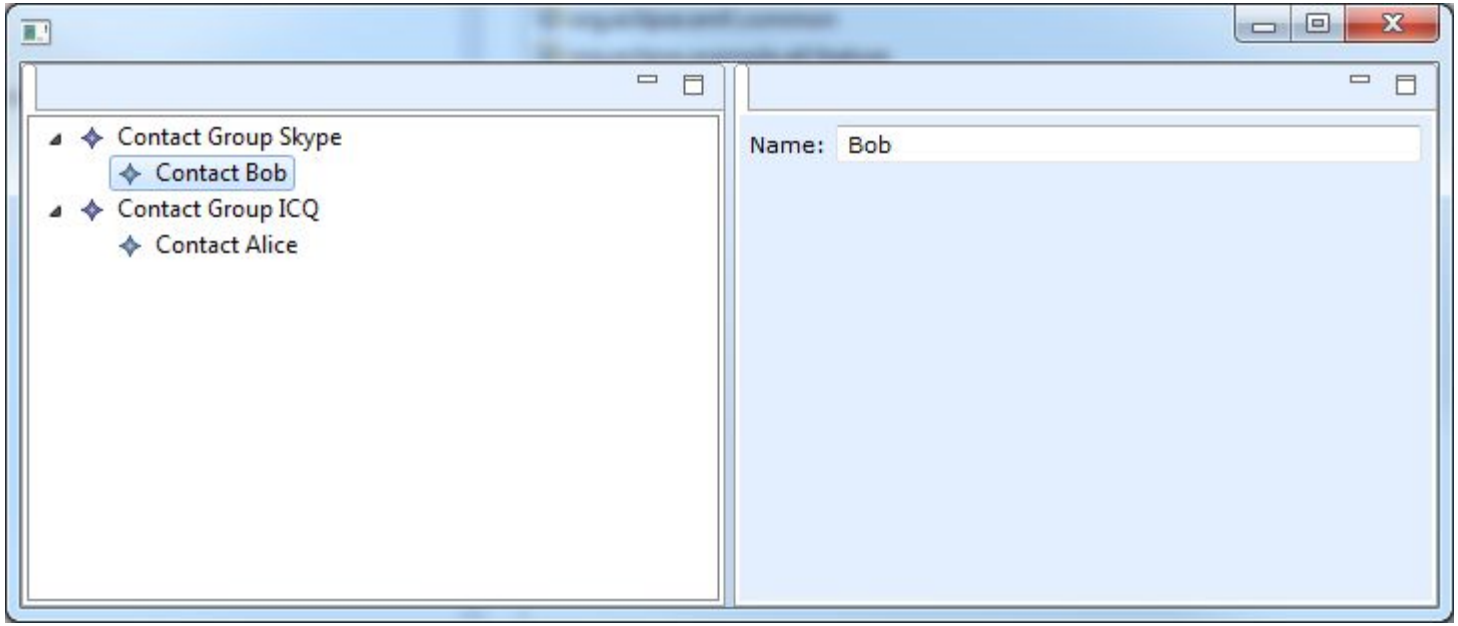
Services

Eclipse 4 includes a lot of services, we will look at one example…

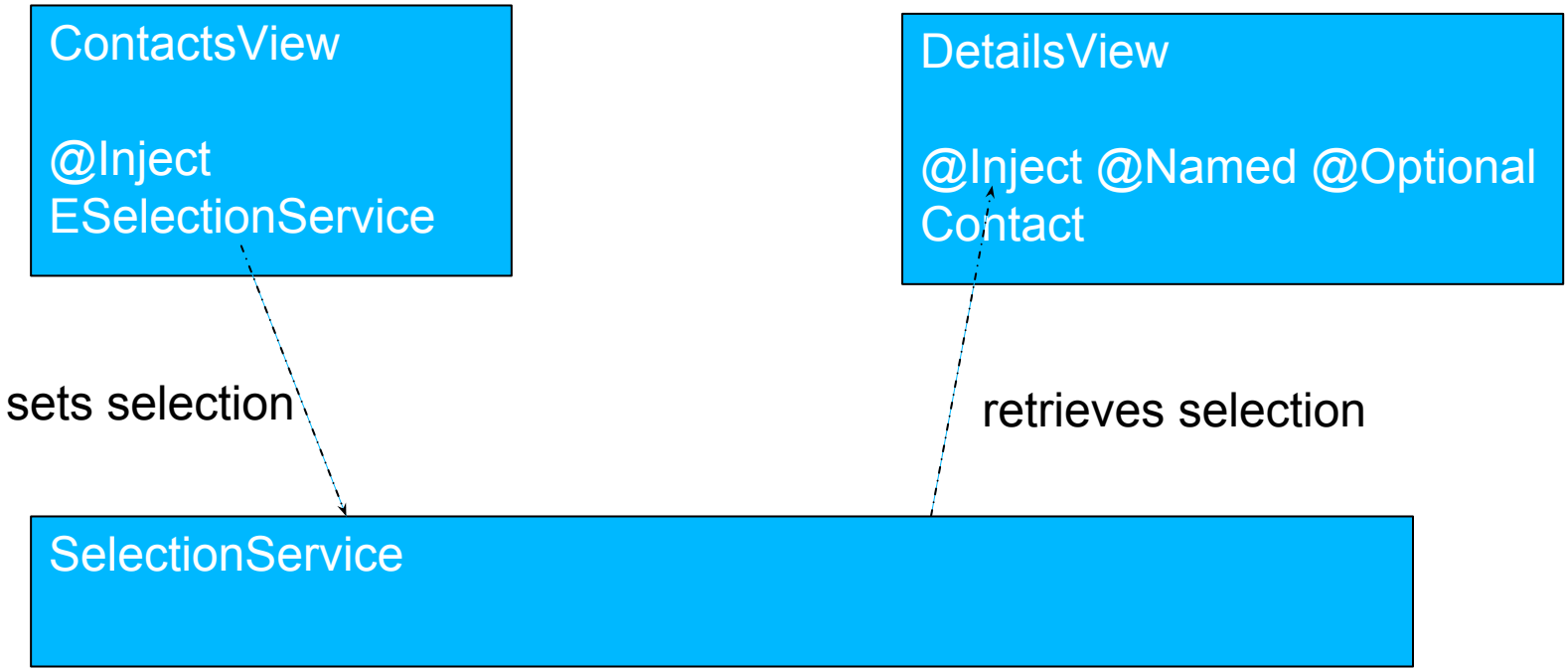
Services


YourClass
@Inject
Service1
OSGi
Service1
Service2
Service3

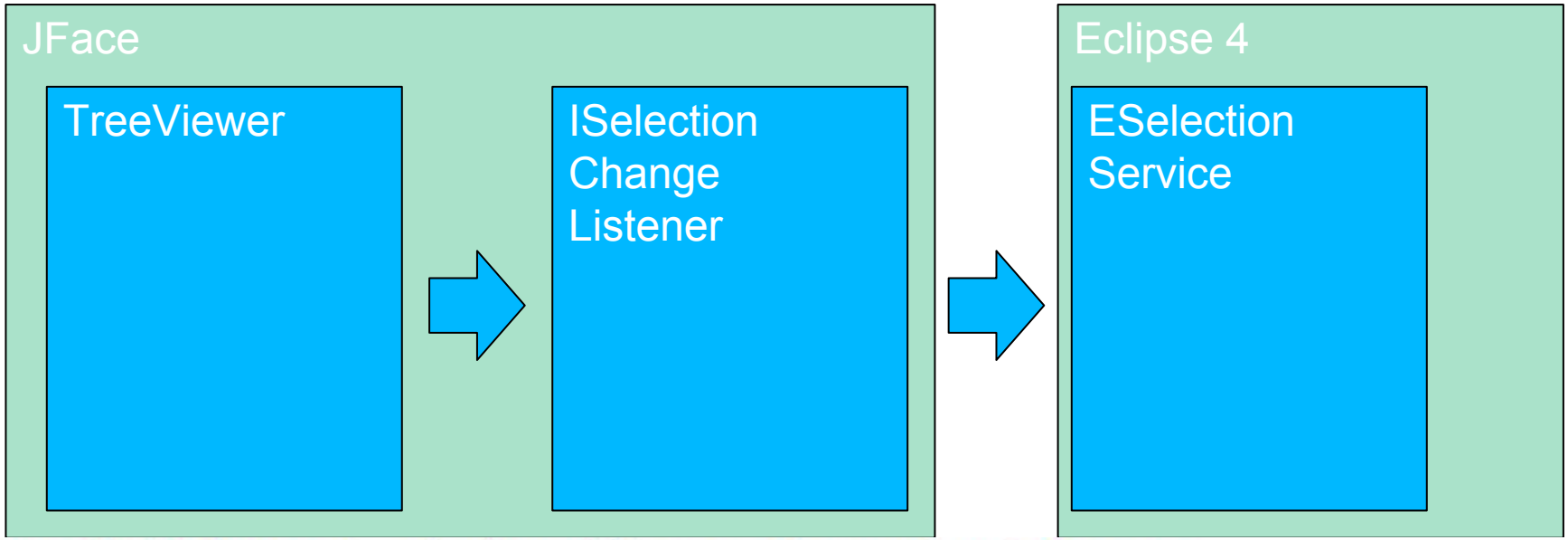
Goal task


Contact Group Skype
Contact Bob
Contact Group ICQ
Contact Alice
Name:
Bob

The selection service



Connecting a Viewer to ESelectionService




```
private final class ISelectionChangedListenerImplementation implements
        ISelectionChangedListener {
    @Override
    public void selectionChanged(SelectionChangedEvent event) {
        Object firstElement = ((IStructuredSelection)event.getSelection()).getFirstElement();
        selectionService.setSelection(firstElement);
    }
}
```

SWTExampleHelper

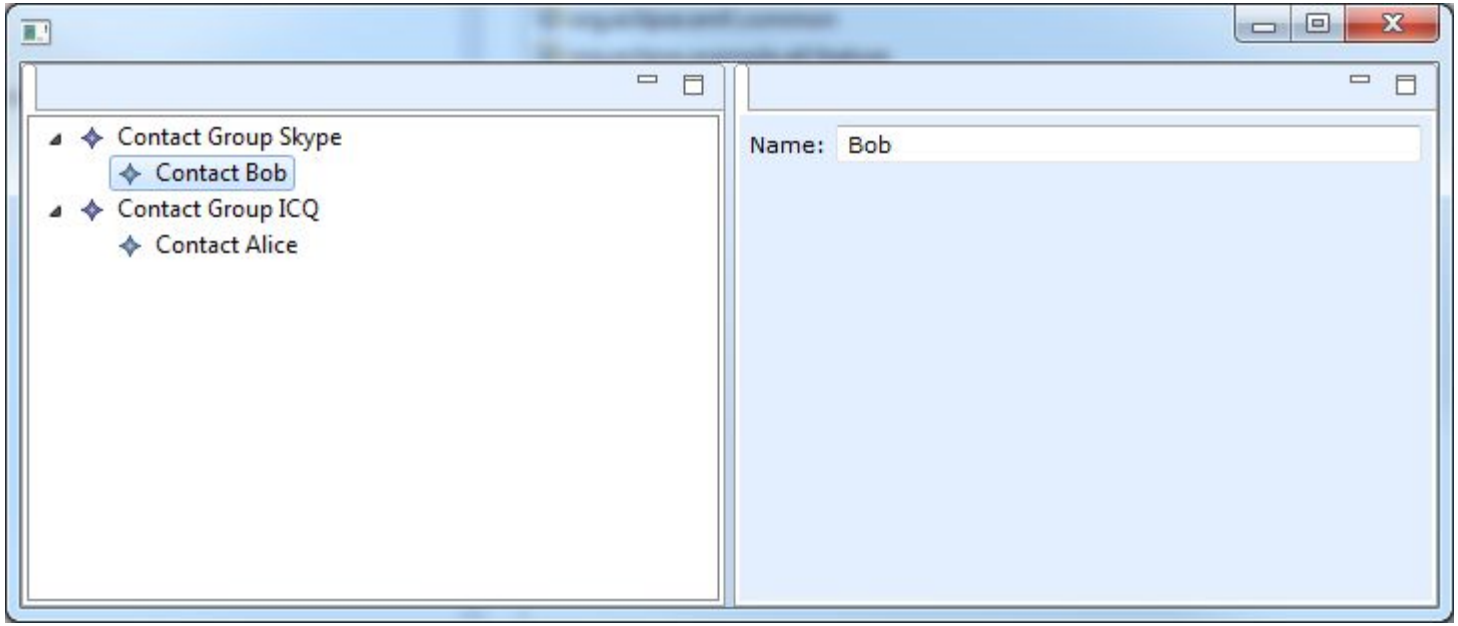
SWTExampleHelper.
connectTreeViewerWithSelectionService
to connect the TreeViewer with the selection service

Inject the Selection

- Use @Inject to get the selection injected
- Use @Optional to accept „null“
- Use @Named to specify that the injected object is the selection
- Use the parameter type to specify the type of selection you want to retrieve

```
@Inject
public void setInput(@Optional
@Named(IServiceConstants.ACTIVE_SELECTION) Contact contact)
```

Goal task 2

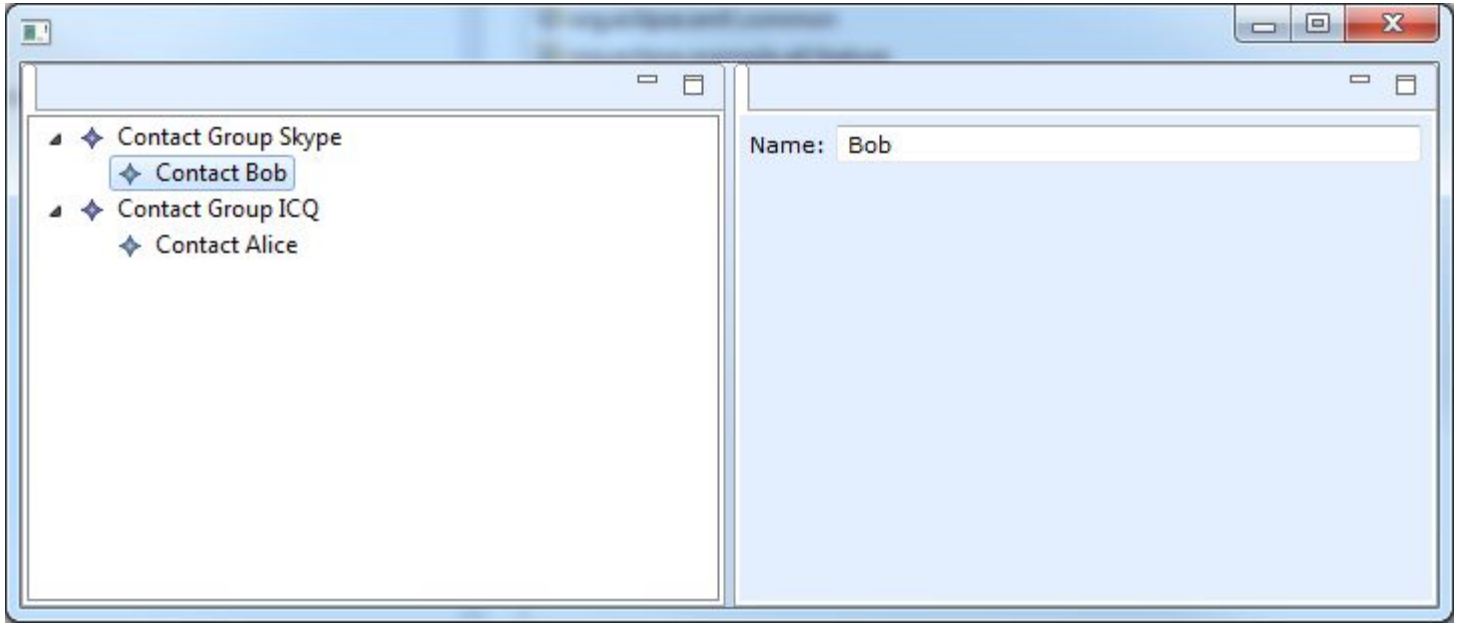

Contact Group Skype
Contact Bob
Contact Group ICQ
Contact Alice
Name:
Bob

Task: Selection Service

- Inject the ESelectionService in the ContactsView an connect it with the TreeViewer (you can use SWTExampleHelper.connectTreeViewer…

- Retrieve the selection in the DetailsView:
  - Create a new method to set the selection
  - Inject the selection using @Inject, @Named and @Optional
  - Constant: IServiceConstants.ACTIVE_SELECTION

Result task 2



```java
public class ContactsView {

    @Inject
    ContactService contactService;

    @Inject
    ESelectionService selectionService;

    private TreeViewer treeViewer;

    @Inject
    public ContactsView(Composite parent) {
        treeViewer = SWTExampleHelper.createTreeViewer(parent);
    }

    @PostConstruct
    public void init() {
        treeViewer.setInput(contactService.getInput());
        SWTExampleHelper.connectTreeViewerWithSelectionService(treeViewer, selectionService);
    }

    @Focus
    public void setFocus(){
        treeViewer.getTree().setFocus();
    }

    @PreDestroy
    public void dispose(){
        SWTExampleHelper.dispose(treeViewer);
    }

}
```

```java
public class DetailsView {

    private Text text;

    @Inject
    public DetailsView(Composite parent) {
        text = SWTExampleHelper.createTextWithLabel(parent);
    }

    @Inject
    public void setInput(
            @Optional @Named(IServiceConstants.ACTIVE_SELECTION) ContactEntry contactEntry) {
        if(contactEntry==null){
            text.setText("");
            return;
        }
        text.setText(contactEntry.getName());
    }

}
```

Outline

- The Application Model
- Implementing Views
- Selection Service

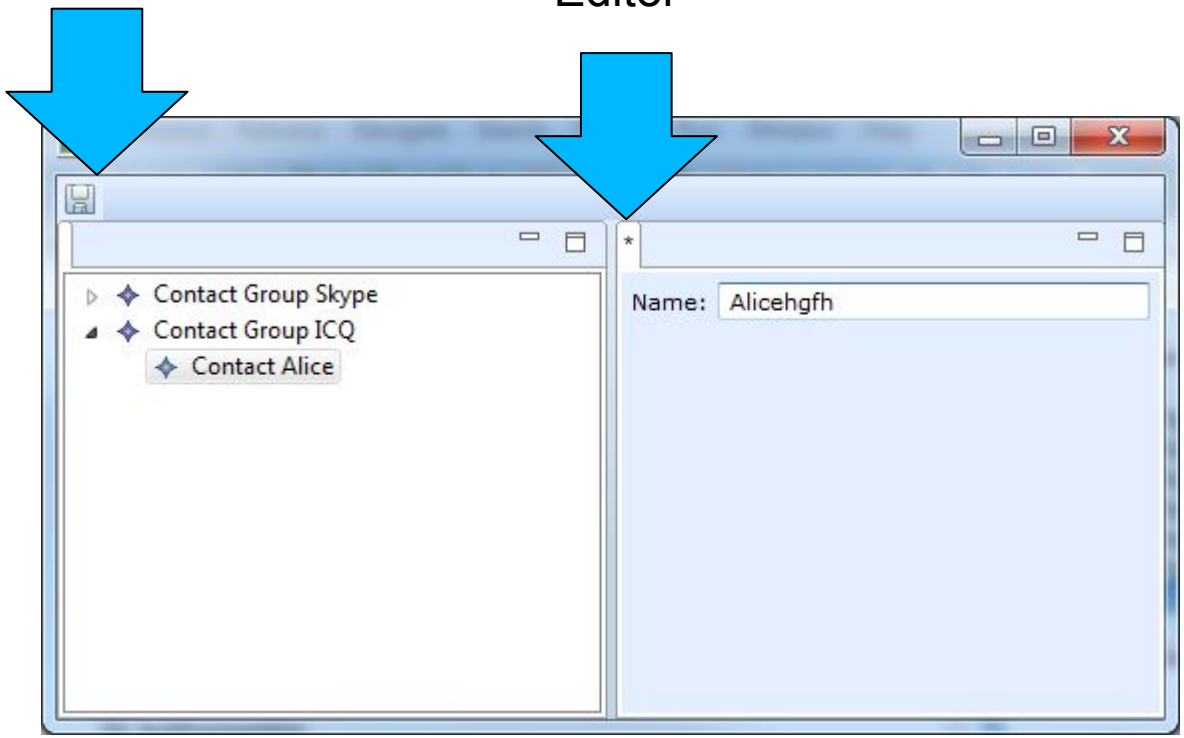
- Handlers, Commands and Items
- Editors

- Dependency Injection Details
- Modularity
- Migration

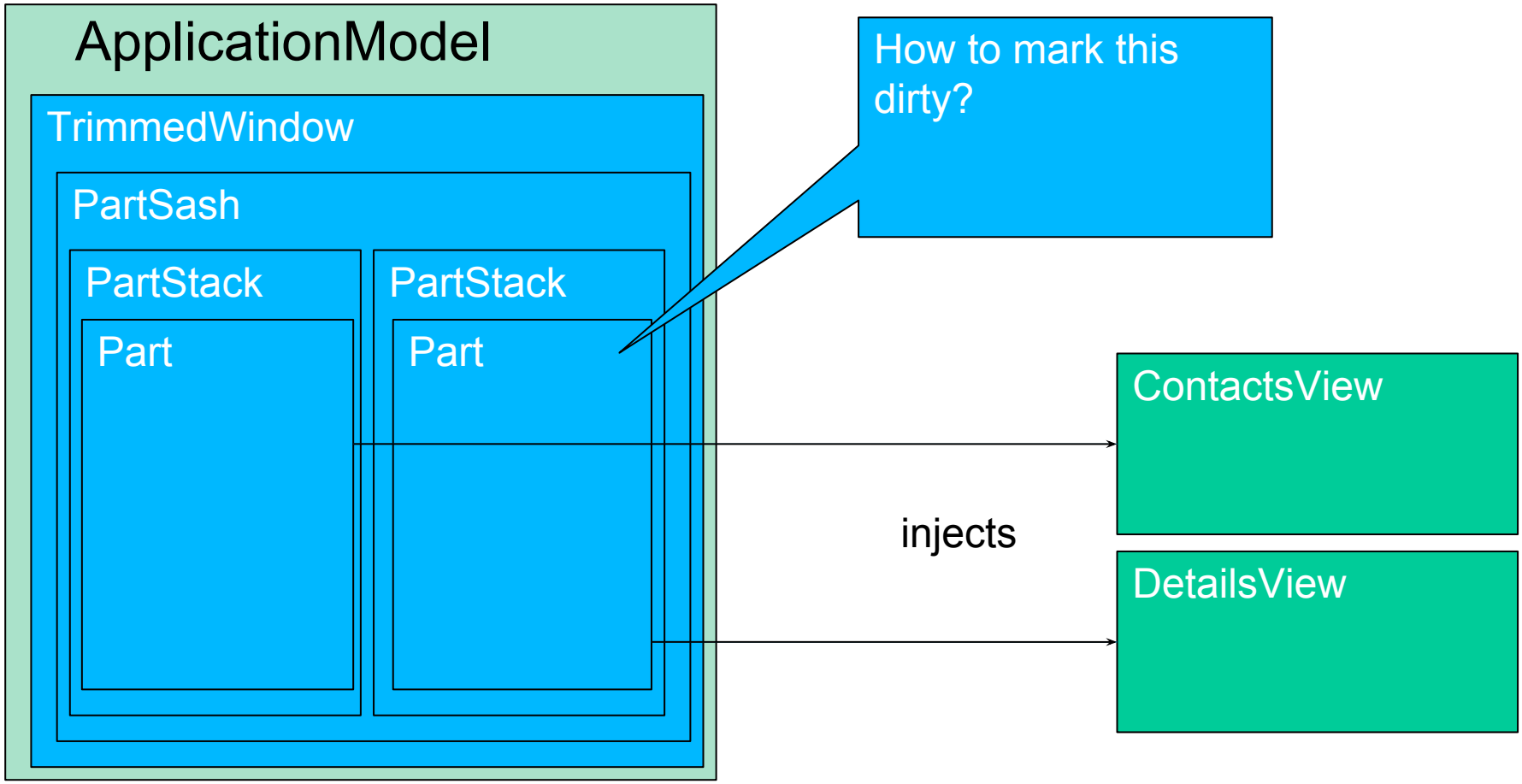
Goal task 3

Save Button

Editor



Application model



Editors in Eclipse 4

- All Parts are of type MDirtyable, get access through:

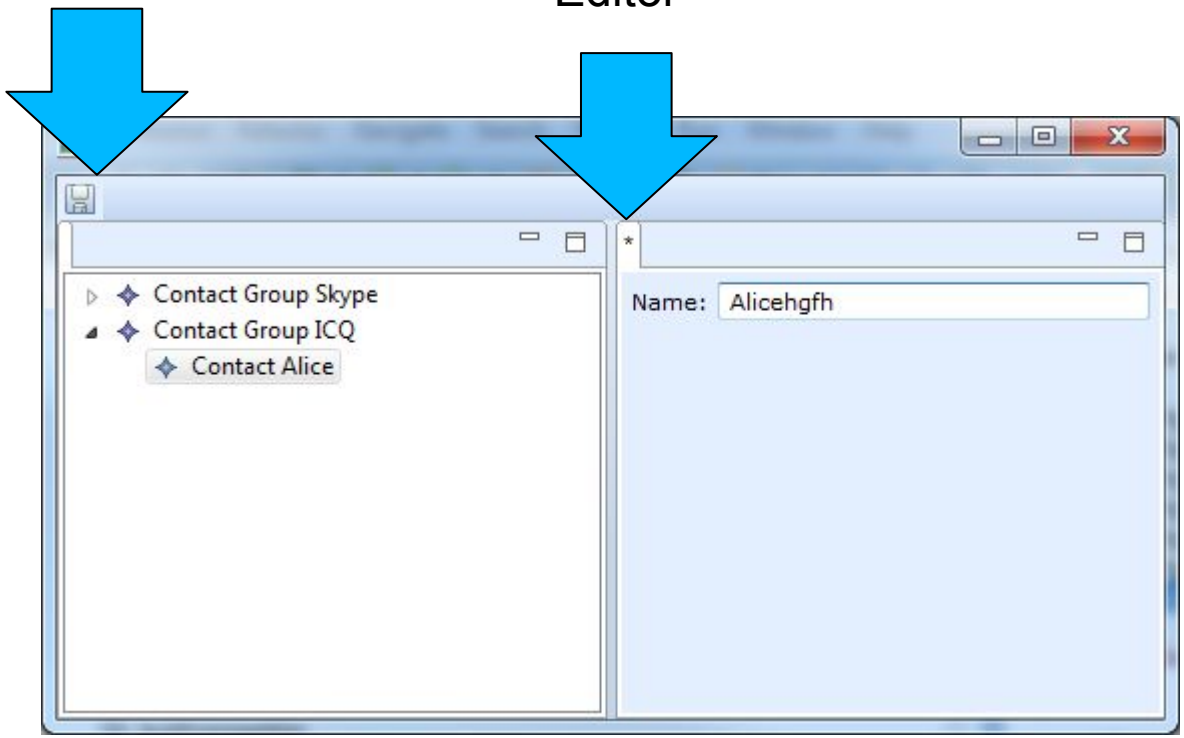
```
@Inject
MDirtyable dirtyable;
dirtyable.setDirty(true/false);
```

- Mark the save method with @Persist
  - In our case in the DetailsView
- You need to set dirty to false, once save is completed

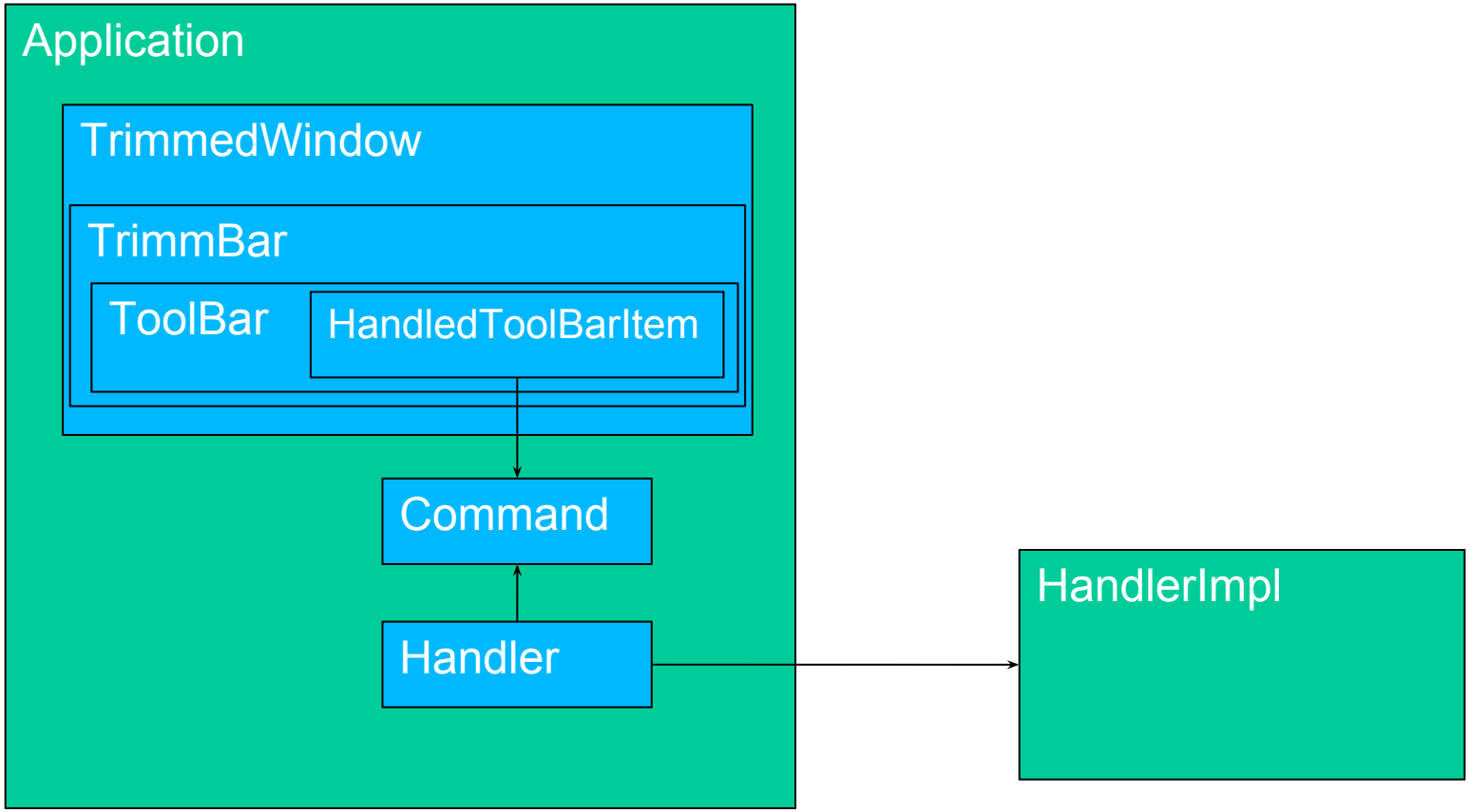
Goal task 3

Save Button

Editor



In e4 handlers, commands and items are part of the workbench model



Handler

- Implementations are POJOs
- They can easily be tested
- Mark the method to be executed with @Execute
- Mark the method which is responsible for the enabled state with @CanExecute
- Needs to return a boolean

- @Execute and @CanExecute include an @Inject, so parameters can be injected, e.g.:
  - Current selection
  - Active part

Save Button in e4

Active part can be injected to check if it is dirty:

```
@CanExecute
public boolean canExecute(@Named(IServiceConstants.ACTIVE_PART)
    @Optional MPart part) {
        //Check for null
        return part.isDirty();
}
```

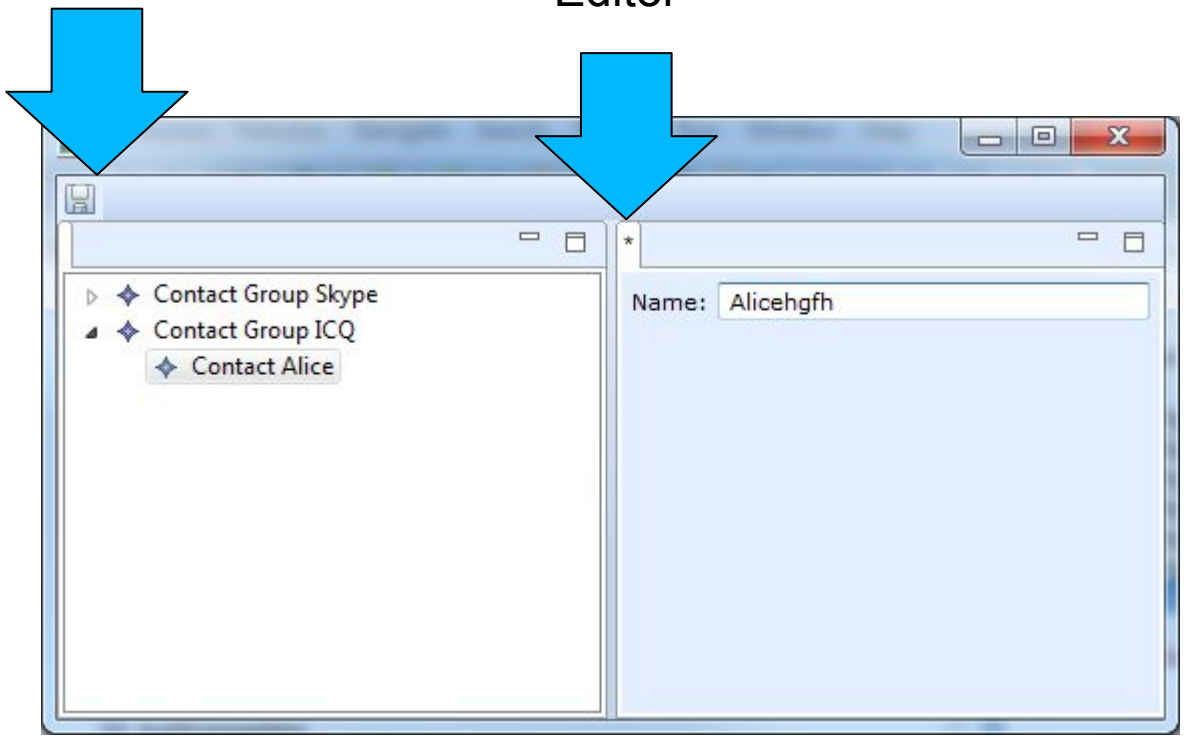
Save can be triggered using the EPartService:

- partService.savePart(part, false);
  - Alternative: partService.saveAll()

Goal task 3

Save Button

Editor

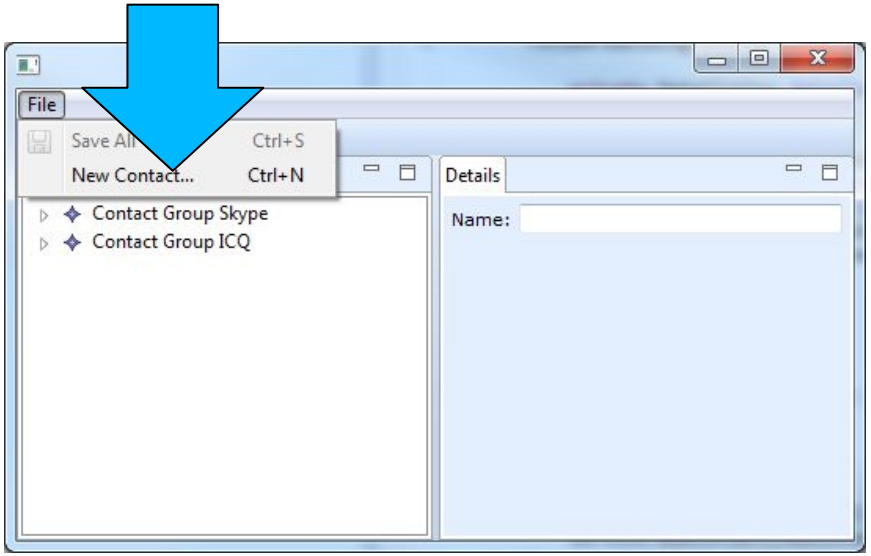


Task: Implement an Editor

- Let the DetailsView handling it‘s dirty state:
  - Set dirty if (Text!=input) on MDirtyable
  - Set not dirty in the save method
- Implement a save method to write changes to the contact. Mark it with @Persist

- Implement a save tool item, saving the active part
  - (@Named(IServiceConstants.ACTIVE_PART) MPart)
- Add a Command, a Handler and an Implementation
- The save tool item should only be enabled, if the active part is dirty

Optional Task



- Only enabled if a ContactGroup is selected
- Reuse the DetailsView in a dialog, e.g. TitleAreaDialog

contact= ModelFactory.eINSTANCE.createContact()
contactGroup.getContacts.add(contact)

DetailsView

```
@Inject
MDirtyable dirtyable;
```

```
@PostConstruct
public void init() {
    text.addModifyListener(new ModifyListener() {

        @Override
        public void modifyText(ModifyEvent e) {
            dirtyable.setDirty(true);

        }
    });
}
```

```
@Persist
public void save() {
    input.setName(text.getText());
    dirtyable.setDirty(false);
}
```

```
@Inject
public void setInput(
        @Optional @Named(IServiceConstants.ACTIVE_SELECTION) ContactEntry contactEntry) {
    if (contactEntry == null) {
        text.setText("");
        input = null;
    } else {
        text.setText(contactEntry.getName());
        input = contactEntry;
    }
    dirtyable.setDirty(false);
}
```

```java
public class SaveHandler {


    @Execute
    public void execute(EPartService partService){
        partService.saveAll(false);
    }

    @CanExecute
    public boolean canExecute(@Named(IServiceConstants.ACTIVE_PART) @Optional MPart part){
        if(part==null){
            return false;
        }
        return part.isDirty();
    }

}
```

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Extending Contexts

IEclipseContext provides methods to browse and modify the Context, e.g.

- context.set(MyObject.class, new MyObject());
- IEclipseContext.modify() walks up in the hierarchy and tries to replace an existing element

- Used to enhance low level contexts, e.g. to pass an input to an editor
- Used to create contexts for testing

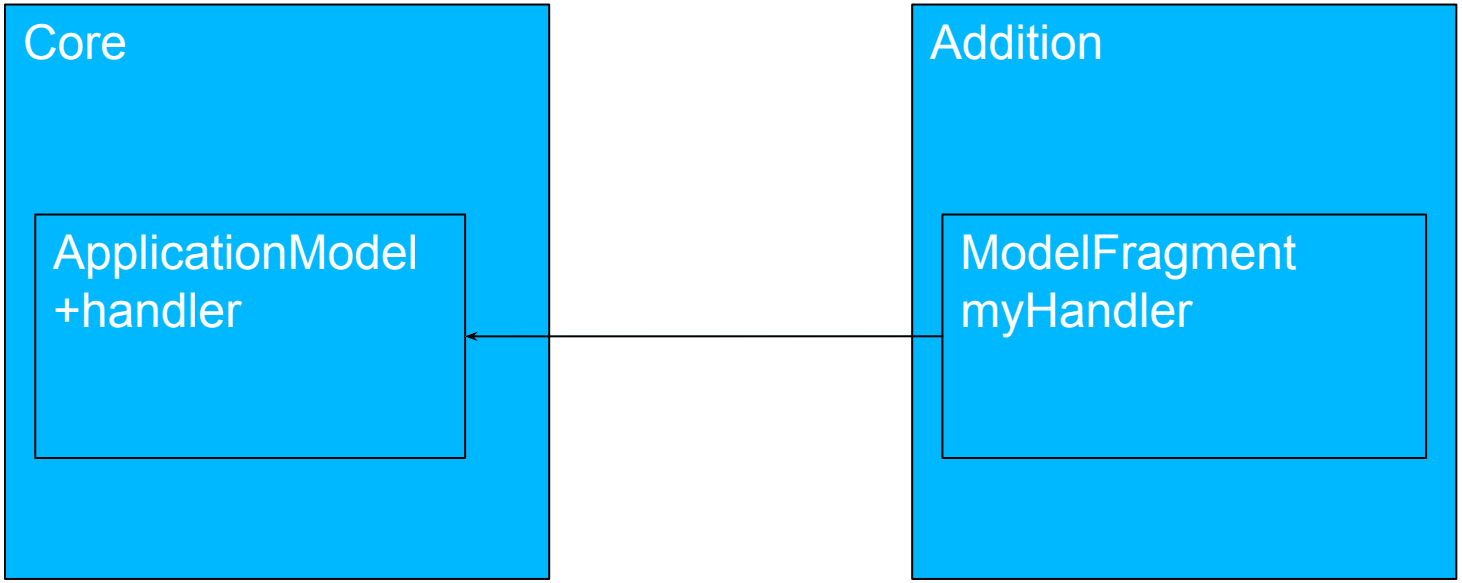
Triggering Dependency Injection

Injection can be manually triggered:

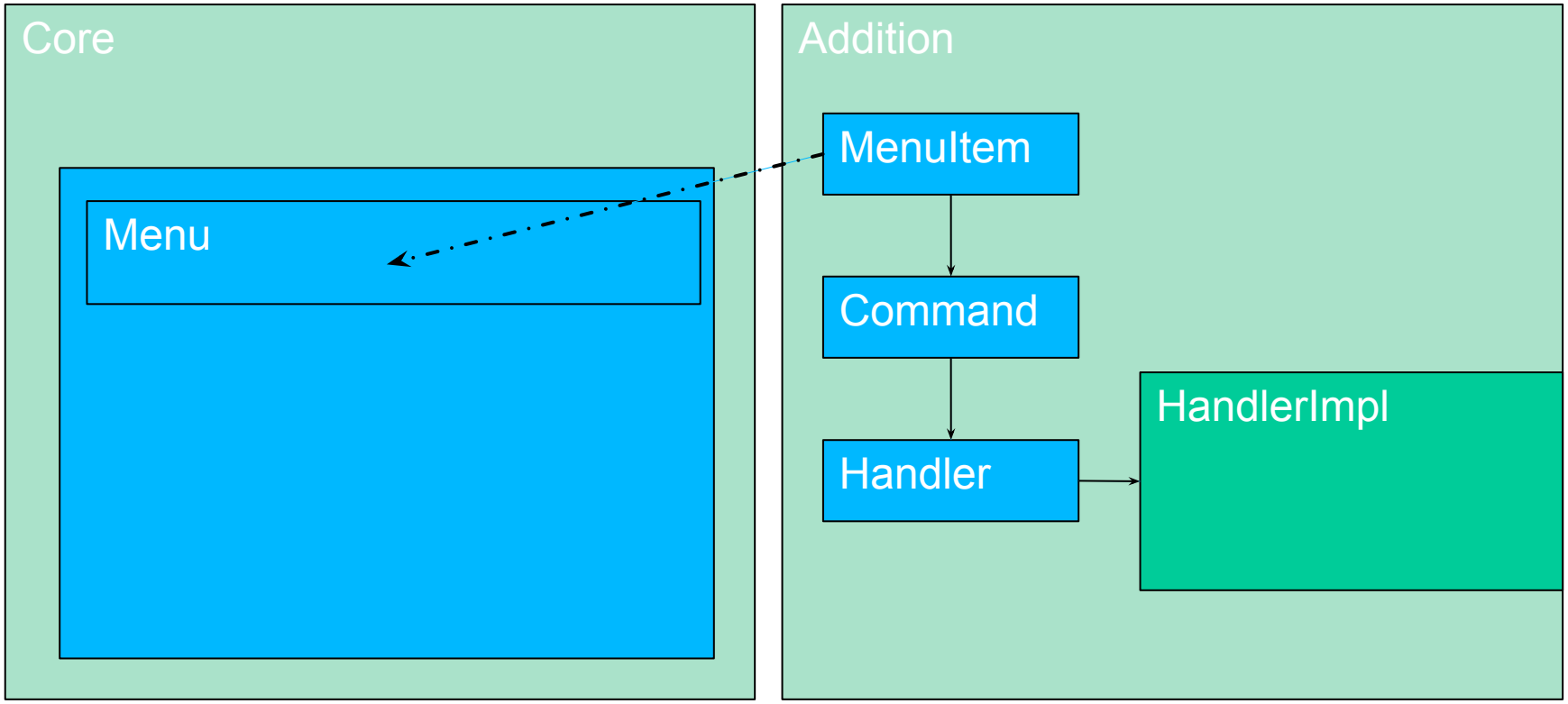
- ContextInjectionFactory.make(DetailsView.class, context);
- Methods with annotation can be called:
- ContextInjectionFactory.invoke(detailsView, Persist.class, context);

Add contributions from other plugins

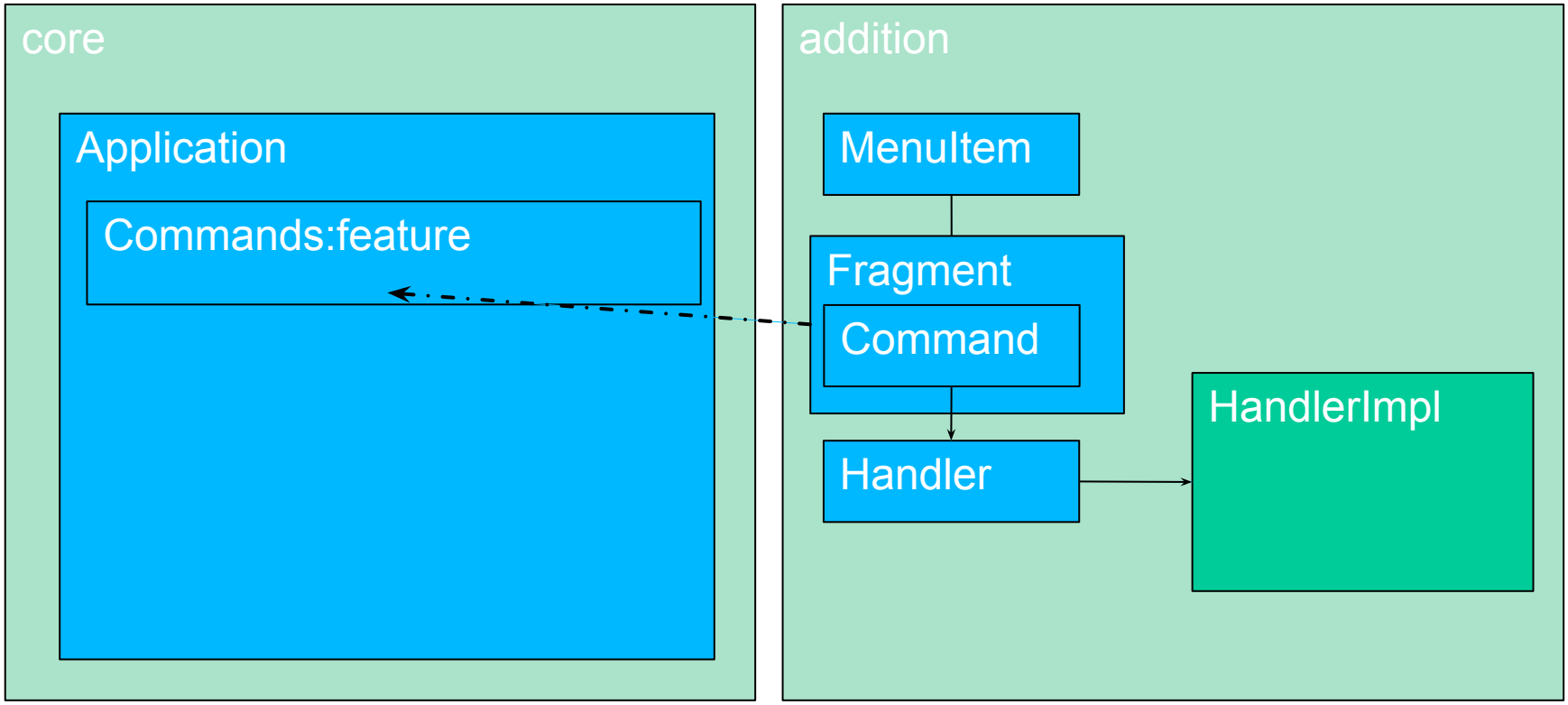
- Contributions can be added from plugins, e.g. to deploy an optional feature



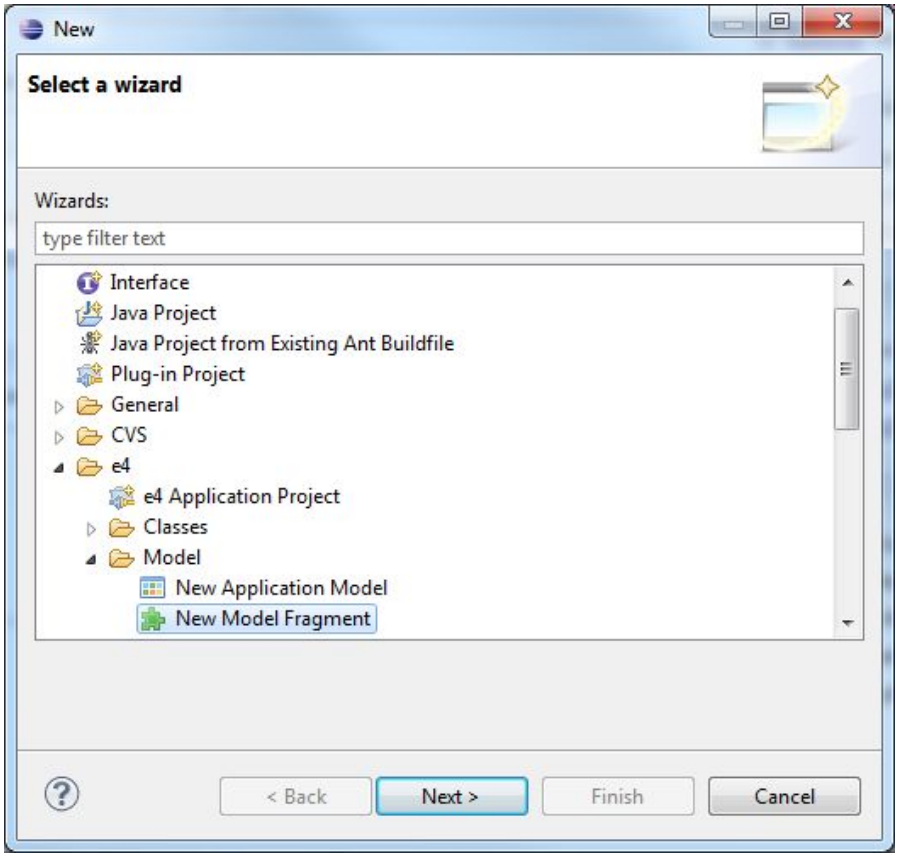
Extend the Application Model



Extend the Application Model

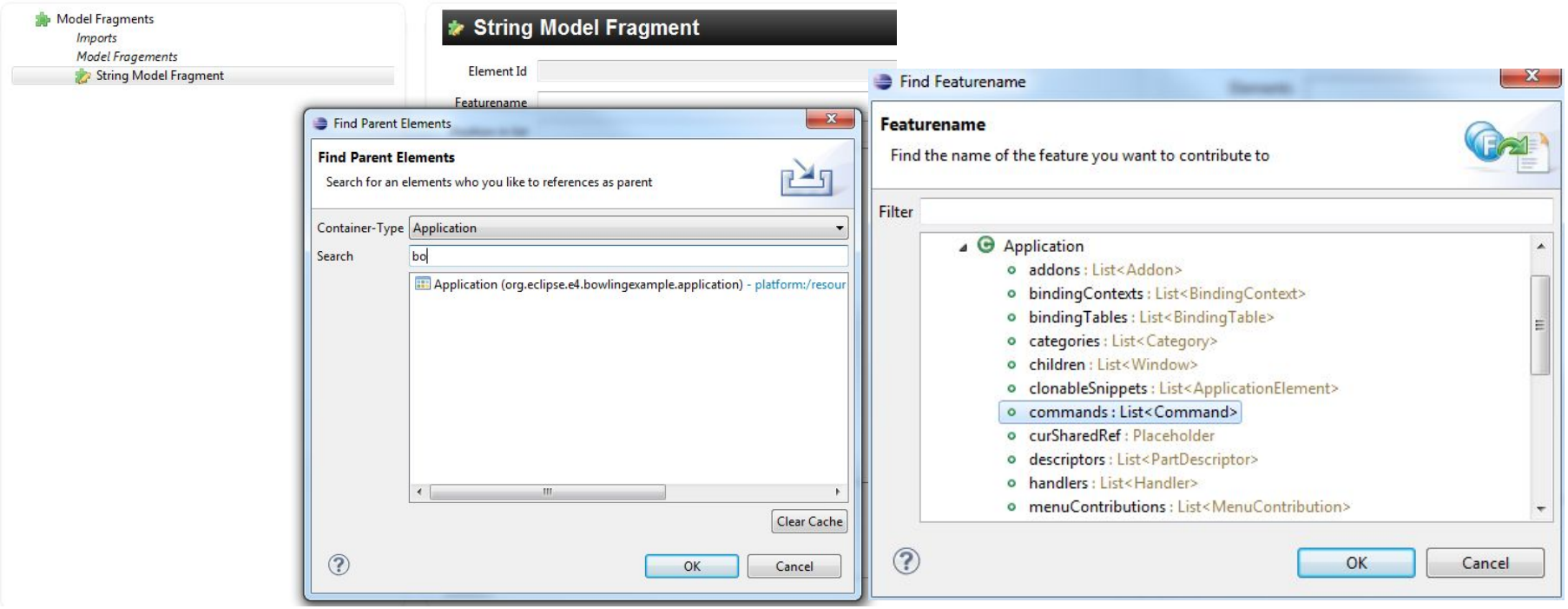


Fragments are files in the workspace

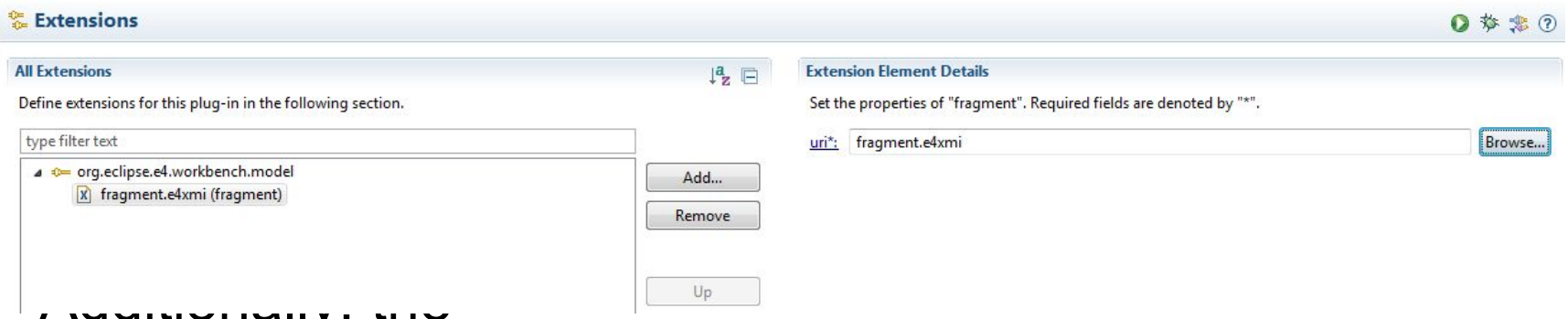


Extend the application model

Extended elements (container) are referenced by ID and EMF features



Fragments need to registered

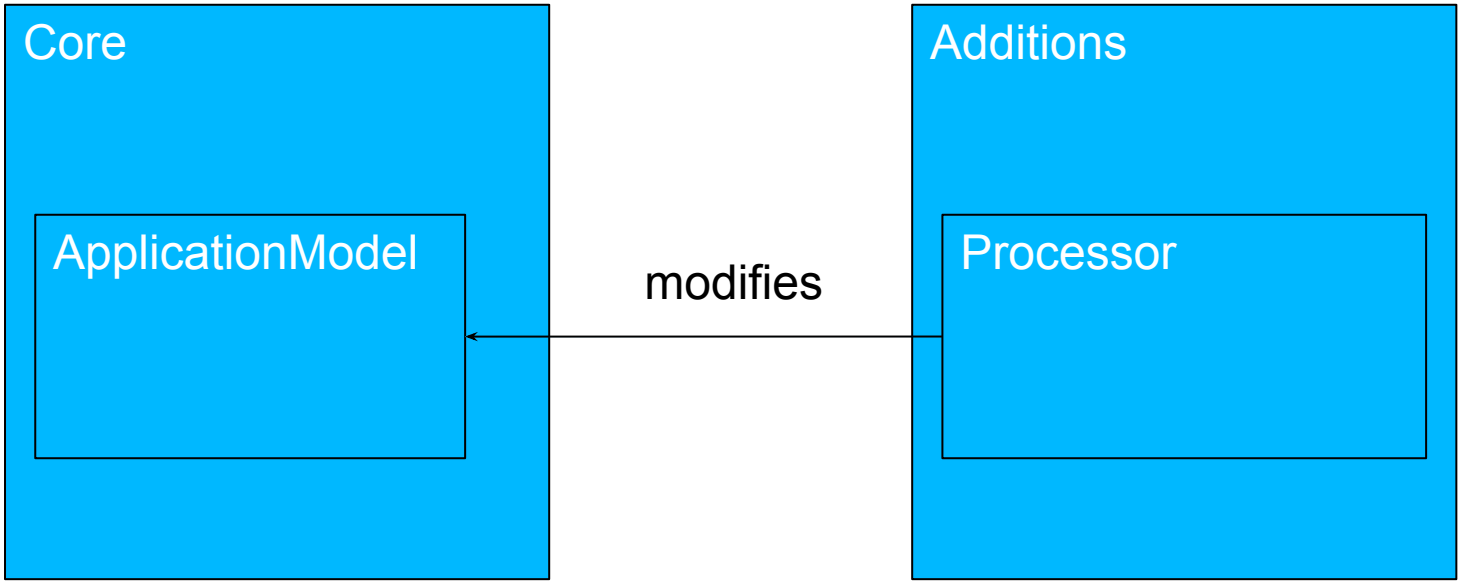


Additionally, the Application Model needs to be registered

```
<extension
      id="product"
      point="org.eclipse.core.runtime.products">
   <product
         application="org.eclipse.e4.ui.workbench.swt.E4Application"
         name="Hyperbola">
      <property
            name="appName"
            value="Hyperbola">
      </property>
      <property
            name="applicationXMI"
            value="org.eclipsercp.e4.hyperbola/Application.e4xmi">
      </property>
   </product>
</extension>
```

Programmatic model enhancements (Optional)

- Processors modify the Application Model using the EMF API



Processors

- Processors need to be registered (like fragments)
- Mark the method to be execute with @Execute

Inject the elements to be extended or modified,

- e.g.: MApplication or EModelService

- Only root context is available

- Use EModelService.findElement() to retrieve Elements

Programmatically modify the application model

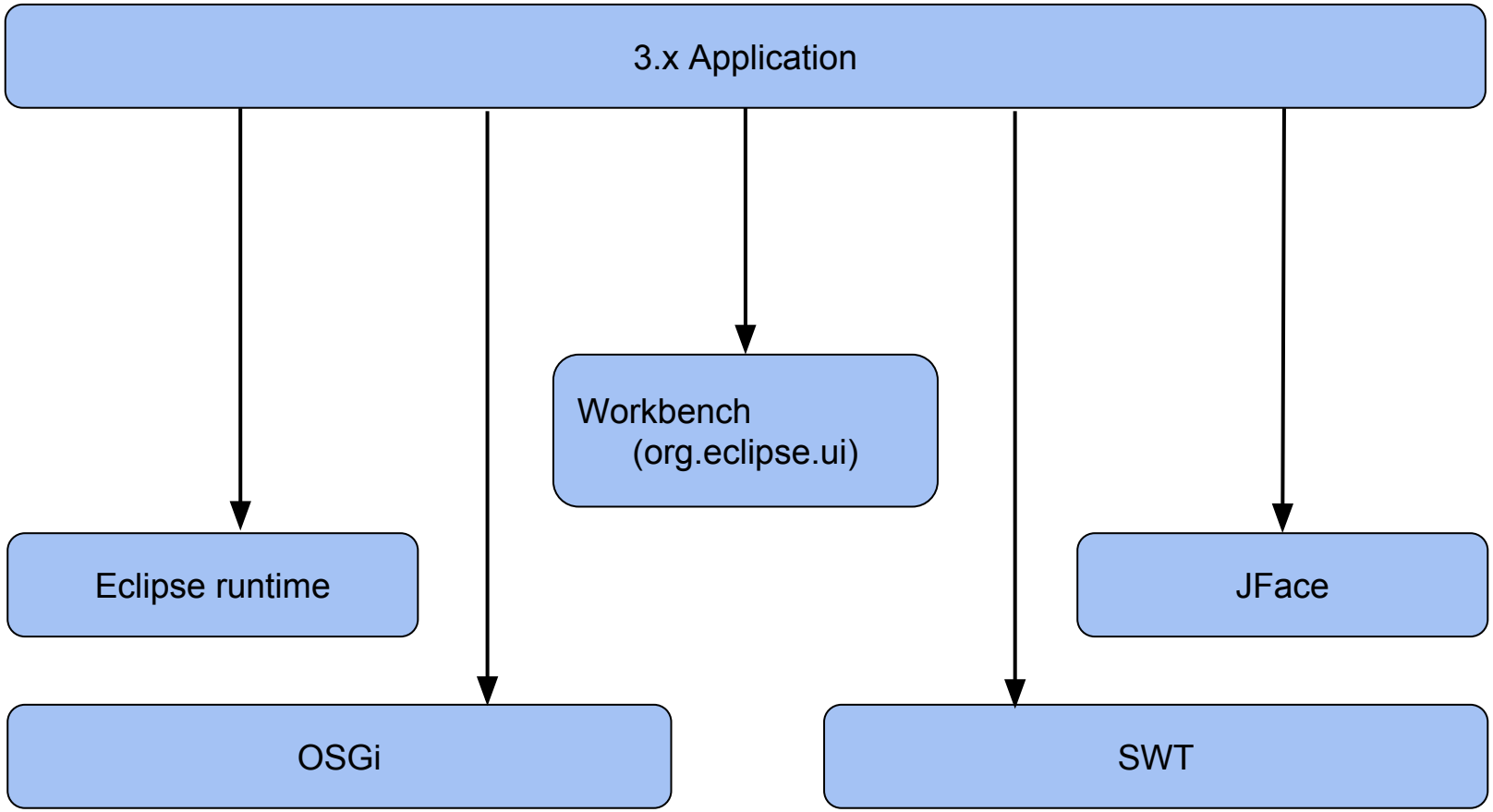
- Model classes are prefixed with a „M“
- They provide getter and setter methods for simple attributes as well as lists for references
- New Elements are created with a factory

```
@Execute
public void execute(MWindow window){
 window.setHeight(200);
}
@Execute
public void execute(MApplication application){
 application.getChildren().add(eModelService.create(MWindow.class))
}
```

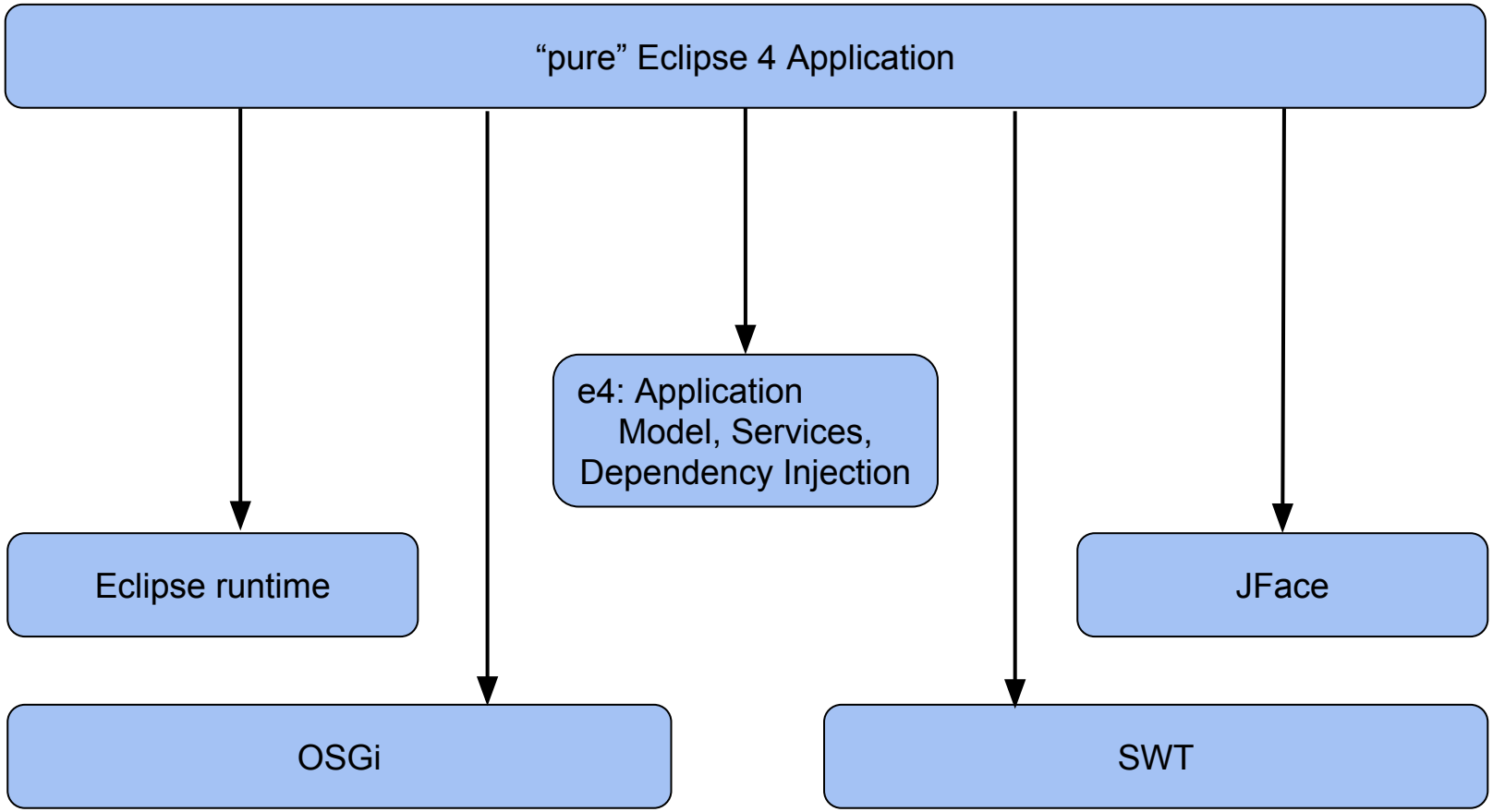
Migration from 3.x?

- E4 offers a compatibility layer
- E4 only offers benefits, if you use the new concepts
- Many things like styling or dependency injection can be used in 3.x, too!

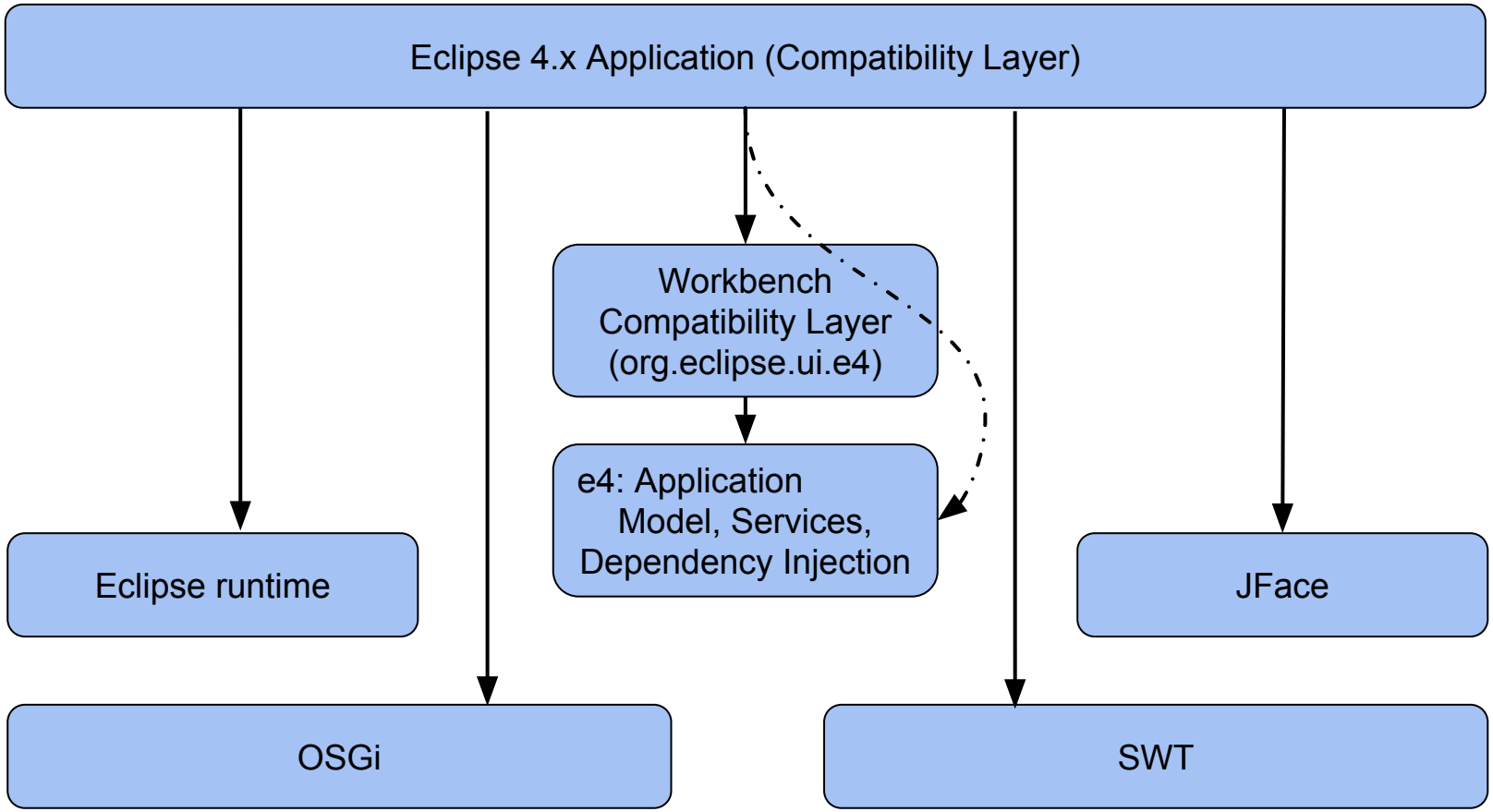
Eclipse 3.x vs. Eclipse 4.x vs. e4?



Eclipse 3.x vs. Eclipse 4 vs. e4?



Eclipse 3.x vs. Eclipse 4 vs. e4?



Eclipse 3.x vs. Eclipse 4 vs. e4?

- Don't go for Eclipse 3.x anymore, choose between:
- "pure" Eclipse 4
- Compatiblity layer (3.x API based on Eclipse 4.x)
- 3.x API and "pure" Eclipse 4 generally do not mix well
- Existing IDE related frameworks usually use the 3.x API
- Look at the frameworks you are going to use:
- Many 3.x based => Go for the compatibility layer
- A few 3.x based => evaluate their adaptation to Eclipse 4
- No 3.x based => go for "pure" Eclipse 4
- If you develop for the 3.x API, do it in an Eclipse 4 friendly way (for more details see online tutorial)

Ways of mixing e4 with 3.x

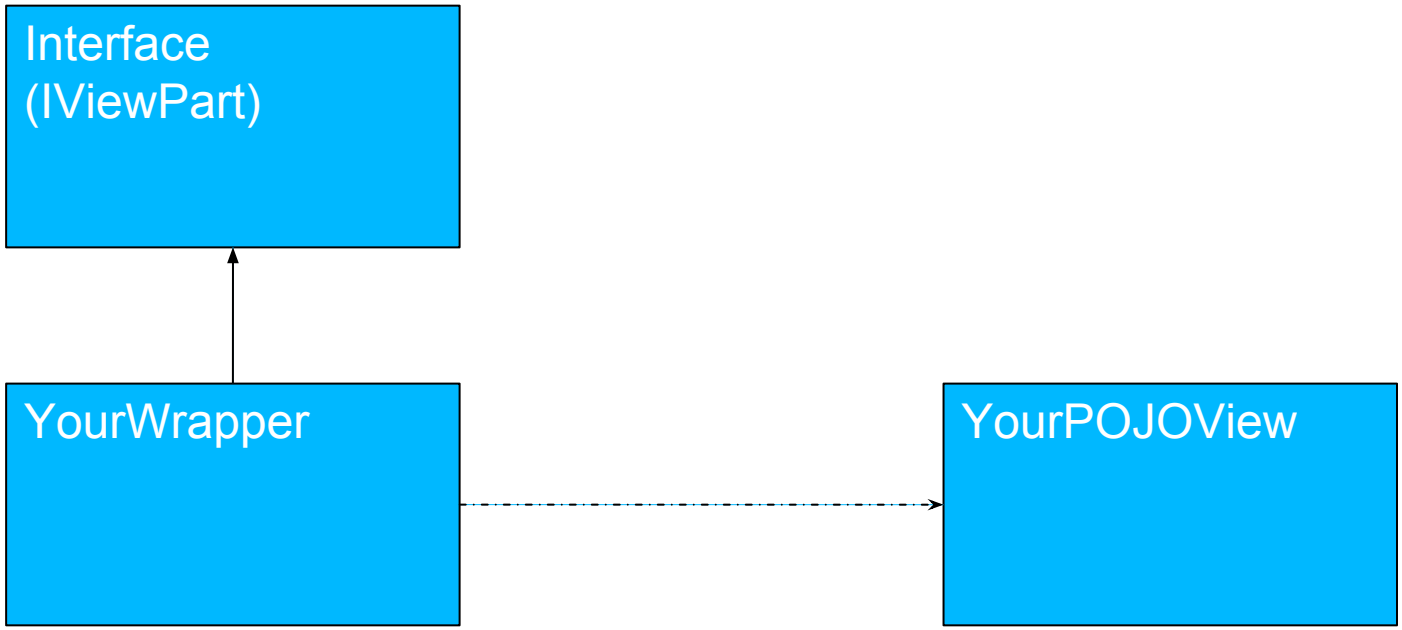
The compatibility layer mocks org.eclipse.ui and translates all calls and extensions to an Application Model in the background

Option 1: Add contributions to this model.

Option 2. Register contributions as in 3.x but use a an e4 POJO (supported for views)

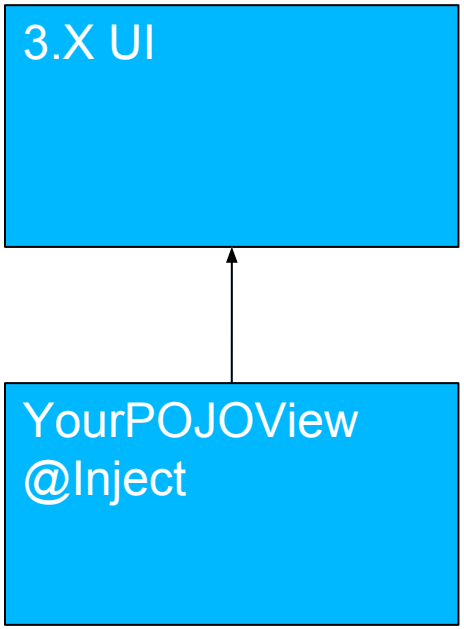
Single Sourcing

Keep your application independant from technologies



Single Sourcing

Since Luna, 3.x layer accepts e4 Views using the view part extension point



More Information

- E4 Wiki
- E4 Newsgroup
- Blogs
- Books
- eclipsesource.com/eclipse4tutorial

- Professional Training => eclipsesource.com
- Jonas.Helming@eclipsesource.com